INCH-POUND
MIL-M-38510/658A
7 MARCH 2006

SUPERSEDING
MIL-M-38510/658
19 December 1986

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH-SPEED CMOS, DECODERS, MONOLITHIC SILICON, POSITIVE LOGIC

Reactivated after 07 Mar. 2006 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS, logic microcircuits. Two product assurance classes and a choice of case outlines, lead finishes, and radiation hardness assurance (RHA) are provided and are reflected in the complete Part or Identifying Number (PIN). For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535 (see 6.3).

1.2 Part or identifying number (PIN). The PIN is in accordance with MIL-PRF-38535 and as specified herein.

1.2.1 Device types. The device types are be as follows:

Device type	Circuit
01	BCD-to-decimal decoder
02	Single 3-to-8 line decoder
03	Dual 2-to-4 line decoder
04	4-to-16 line decoder/demultiplexer
52	Single 3-to-8 line decoder (TTL compatible)
53	Dual 2-to-4 line decoder (TTL compatible)

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
K	GDFP2-F24 or CDFP3-F24	24	Flat pack
L	GDFP3-F24 or CDFP4-F24	24	Dual-in-line
2	CQCC1-N20	20	Square leadless chip carrier
3	CQCC1-N28	28	Square leadless chip carrier

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or email CMOS@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

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1.3 Absolute maximum ratings.

Supply voltage range (V_{CC})	-0.5 V dc to +7.0 V dc
DC input voltage range (V_{IN})	-0.5 V dc to V_{CC} + 0.5 V dc
DC output voltage range (V_{OUT})	-0.5 V dc to V_{CC} + 0.5 V dc
Clamp diode current (I_{IK}, I_{OK})	±20 mA
DC output current per pin (I_{OUT})	±25 mA
DC V_{CC} or GND current per pin (I_{OC})	±50 mA
Storage temperature range (T_{STG})	-65° to +150°C
Maximum power dissipation, (P_D)	300 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC})	See MIL-STD-1835
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Device types 01, 02, 03, and 04:

Supply voltage range (V_{CC})	2.0 V dc to 6.0 V dc
Maximum input low voltage (V_{IL})	0.3 V at V_{CC} = 2.0 V 0.9 V at V_{CC} = 4.5 V 1.2 V at V_{CC} = 6.0 V
Minimum input high voltage (V_{IH})	1.5 V at V_{CC} = 2.0 V 3.15 V at V_{CC} = 4.5 V 4.2 V at V_{CC} = 6.0 V
Maximum input rise and fall times (t_r, t_f):	
V_{CC} = 2.0 V	1000 ns
V_{CC} = 4.5 V	500 ns
V_{CC} = 6.0 V	400 ns

Device types 52 and 53:

Supply voltage range (V_{CC})	4.5 V dc to 5.5 V dc
Maximum input low voltage (V_{IL})	0.8 V at V_{CC} = 4.5 V through 5.5 V
Minimum input high voltage (V_{IH})	2.0 V at V_{CC} = 4.5 V through 5.5 V
Maximum input rise and fall times (t_r, t_f):	
V_{CC} = 4.5 V and 5.5 V	500 ns

All devices:

Output voltage (V_{OUT})	0.0 V to V_{CC}
Case operating temperature range (T_C)	-55° to +125°C

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard Microcircuits.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.3.2 Logic diagram. The logic diagram shall be as specified on figure 2.

3.3.3 Truth tables. The truth tables and logic equations shall be as specified on figure 3.

3.3.4 Switching time test circuit and waveforms. The switching time test circuit and waveforms shall be as specified on figure 4.

3.3.5 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity or preparing activity upon request.

3.3.6 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 39 (see MIL-PRF-38535, appendix A).

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4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in method 1015 of MIL-STD-883.

b. Delete the sequence specified as interim (pre-burn-in) electrical parameters through interim (post-burn-in) electrical parameters of method 5004 and substitute lines 1 through 7 of table II herein.

c. Burn-in (method 1015 of MIL-STD-883).

(1) Unless otherwise specified in the manufacturers QM plan for static tests (test condition A), ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

i. For static burn-in I, all inputs shall be connected to GND. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs and required on outputs and required on outputs connected to $V_{CC}/2$. R = 680Ω to 47 kΩ.

ii. For static burn-in II, all inputs shall be connected through the resistors to V_{CC}. Outputs shall be open or connected to $V_{CC}/2$. Resistors are optional on outputs if open. Resistors are required on inputs and required on outputs connected to $V_{CC}/2$. R = 680Ω to 47 kΩ.

iii. $V_{CC}/2 = V_{CC}/2 \pm 0.5$ V.

iv. V_{CC} = 6.0 V ±0.5 V for device types 01-04.
V_{CC} = 5.5 V ±0.5 V for device types 52 and 53.

(2) Unless otherwise specified in the manufacturers QM plan for dynamic test (test condition D), ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015 of MIL-STD-883.

i. For dynamic burn-in, all select inputs shall be connected through the resistors R1 in parallel to the clock pulses as shown in paragraph v below. All outputs shall be connected through the resistors R2 to $V_{CC}/2$. Output resistors = 1 kΩ ±5%.

ii. V_{CC} = 6.0 V ±0.5 V for device types 01-04; V_{CC} = 5.5 V ±0.5V for device types 52 and 53.

iii. $V_{CC}/2 = V_{CC}/2 \pm 0.5$ V.

iv. Duty cycle = 50 percent ±15percent; V_{IH} = 4.5 V minimum to V_{CC} maximum; V_{IL} = 0.0 V ±0.5V; t_{TLH} and $t_{THL} \leq 500$ ns.

v. For device type 01, input A, CP1 = 100 kHz square wave; input B, CP2 = 50 kHz square wave; input C, CP3 = 25 kHz square wave; input D, CP4 = 12.5 kHz square wave.

For device types 02 and 52, inputs A, B, and C, CP1, CP2, and CP3 = 25 kHz to 1 MHz square wave; CP3 = CP2/2, CP2 = CP1/2. Enable inputs $\overline{G2A}$ and $\overline{G2B}$ are connected to GND and G1 is connected through the resistor R2 to $V_{CC}/2$.

For device types 03 and 53, input A, CP1 = 25 kHz to 1 MHz square wave; input B, CP2 = CP1/2. Enable input G is connected to GND.

For device type 04, inputs A, B, C, D, CP1, CP2, CP3, CP4 = 25 kHz to 1 MHz square wave; CP2 = CP1/2, CP3 = CP2/2, CP4 = CP3/2. Enable inputs G1 and G2 are connected through the resistors R1 to GND.

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d. Interim and final electrical test parameters shall be as specified in table II.

e. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failure shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-PRF-38535 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 shall be omitted.

c. Subgroup 4 (C_{IN} measurement) shall be measured only for initial qualification and after process or design changes that may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
High level output voltage	V_{OH1} 2/	V_{IH} = 1.5 V, V_{IL} = 0.3 V I_{OH} = -20 µA	01,02 03,04	2.0V	1.95		V
	V_{OH2} 2/	V_{IH} = 3.15 V, V_{IL} = 0.9 V I_{OH} = -20 µA	01,02 03,04	4.5V	4.45		V
	V_{OH3}	V_{IH} = 4.2 V, V_{IL} = 1.2 V I_{OH} = -20 µA	01,02 03,04	6.0V	5.95		V
	V_{OH4} 2/	V_{IH} = 3.15 V; V_{IL} = 0.9 V I_{OH} = -4.0 mA	01,02 03,04	4.5V	3.7		V
	V_{OH5}	V_{IH} = 4.2 V, V_{IL} = 1.2 V I_{OH} = -5.2 mA	01,02 03,04	6.0V	5.2		V
	V_{OH6}	V_{IH} = 2.0 V, V_{IL} = 0.8 V I_{OH} = -20 µA	52,53	4.5V	4.40		V
	V_{OH7}	V_{IH} = 2.0 V, V_{IL} = 0.8 V I_{OH} = -4.0 mA	52,53	5.5V	3.70		V
Low level output voltage	V_{OL1} 2/	V_{IL} = 0.3 V, V_{IH} = 1.5 V I_{OL} = 20 µA	01,02 03,04	2.0V		0.05	V
	V_{OL2} 2/	V_{IL} = 0.9 V, V_{IH} = 3.15 V I_{OL} = 20 µA	01,02 03,04	4.5V		0.05	V
	V_{OL3}	V_{IL} = 1.2 V, V_{IH} = 4.2 V I_{OL} = 20 µA	01,02 03,04	6.0V		0.05	V
	V_{OL4} 2/	V_{IL} = 0.9 V, V_{IH} = 3.15 V I_{OL} = 4.0 mA	01,02 03,04	4.5V		0.4	V
	V_{OL5}	V_{IL} = 1.2 V, V_{IH} = 4.2 V I_{OL} = 5.2 mA	01,02 03,04	6.0V		0.4	V
	V_{OL6}	V_{IL} = 0.8 V, V_{IH} = 2.0 V I_{OL} = 20 µA	52,53	4.5V		0.05	V
	V_{OL7}	V_{IL} = 0.8 V, V_{IH} = 2.0 V I_{OL} = 4.0 mA	52,53	5.5V		0.4	V
Positive input clamp voltage	V_{IC}+	I_{IN} = 1 mA T_C = +25°C	All	GND		1.5	V
Negative input clamp voltage	V_{IC}-	I_{IN} = -1 mA T_C = +25°C	All	OPEN		-1.5	V
Input current low	I_{IL}	V_{IN} = GND	01,02 03,04	6.0V		-0.1	µA
			52,53	5.5V		-0.1	µA
Input current high	I_{IH}	V_{IN} = V_{CC}	01,02 03,04	6.0V		0.1	µA
			52,53	5.5V		0.1	µA

See footnotes at the end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions 1/ T_C = +125°C unless otherwise specified	Device type	V_{CC}	Limits Min	Limits Max	Unit
Short-circuit output current	I_{OS1} 2/	T_C = -55°C to +125°C	01,02 03,04	2.0V	-2	-50	mA
	I_{OS2} 2/	V_O = GND		4.5V	-15	-150	
	I_{OS3} 2/	V_I = GND		6.0V	-25	-180	
	I_{OS4}			4.0V	-10	-120	
	I_{OS2} 2/		52,53	4.5V	-15	-150	
	I_{OS5} 2/			5.5V	-25	-180	
	I_{OS4}			4.0V	-10	-120	
Supply current quiescent	I_{CC}	V_I = GND or V_{CC}	All	6.0V		20	µA
Additional supply current quiescent per input pin (one unit load)	$I_{CC}\Delta$ 3/	V_{IL} = 0.8 V V_{IH} = 2.4 V Test pin at V_{IN} = 2.4 V Other pins at 0.8 V or 5.5 V I_O = 0 V	52	5.5V		3.0	mA
			53			1.5	
Input capacitance	C_I	T_C = +25°C	All			10	pF
Power dissipation capacitance	C_{PD} 2/ 4/	T_C = +25°C	01			65	pF
			02			85	
			03			75	
			04			90	
			52			85	
			53			75	
Propagation delay times, input to output 5/ 6/	t_{PHL1}, t_{PLH1}	C_L = 50 pF ± 10%	01	4.5V	4	35	ns
	t_{PLH1}		02	4.5V	4	47	
	t_{PHL1}				5		
	t_{PLH1}, t_{PHL1} 4 levels		03	4.5V	6	51	
	t_{PLH1}, t_{PHL1} 5 levels				8		
	t_{PLH1}, t_{PHL1}		04	4.5V	6	44	
	t_{PLH1}		52	4.5V	5	47	
	t_{PHL1}				6		
	t_{PLH1}, t_{PHL1} 4 levels		53	4.5V	5	41	
	t_{PLH1}, t_{PHL1} 5 levels				7	41	
Propagation delay times, enable to output 5/ 6/	t_{PHL2}, t_{PLH2}	C_L = 50 pF ± 10%	02	4.5V	4	41	ns
			03	4.5V	6	41	
			04	4.5V	6	42	
			52	4.5V	4	47	
			53	4.5V	5	47	
Transition delay times 5/ 6/	t_{THL}, t_{TLH}		All	4.5V	3	20	

1/ Complete terminal conditions and limits shall be as specified in table III.

2/ Guaranteed but not tested.

3/ Total supply current = I_{CC} + $I_{CC}\Delta$.

4/ Power dissipation capacitance (C_{PD}) per decoder/demultiplexer.

5/ Tested at V_{CC} = 4.5 V at +125°C for sample testing and V_{CC} = 4.5 V at +25°C for screening. Guaranteed at other V_{CC} voltages and temperatures, see table IA and exception in 4.4.1d.

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TABLE I. Electrical performance characteristics – Continued.

6/ For propagation and transition delay times at V_{CC} = 2.0 V, increase limit by a factor of 5.
For propagation and transition delay times at V_{CC} = 6.0 V, decrease limit by a factor of .85.

TABLE IA. Calculated dynamic figures at -55°C and +25°C ambient temperatures.

V_{CC} 1/	T_A = (°C) 2/	
	+125	-55 and +25
2.0 V	5 ↑	5 x 0.75 ↑
4.5 V	1 →	→ 0.75 ↓
6.0 V	0.85	0.85 x 0.75

1/ The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value. Rounding off according 5/4.

2/ Normalized numbers (+125°C equals 1).

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TABLE II. Electrical test requirements.

Line no.	MIL-PRF-38535 test requirements	Class S device 1/			Class B device 1/		
		Ref. par.	Table III Subgroups 2/	Table IV delta limits 3/	Ref. par.	Table III subgroups 2/	Table IV delta limits 3/
1	Interim electrical parameters		1			1	
2	Static burn-in I (method 1015)	4.2c 4.5.2	Req'd			Not req'd	
3	Same as line 1		1*	Δ			
4	Static burn-in II (method 1015)	4.2c 4.5.2	Req'd		4.2c 4.5.2	Req'd 4/	
5	Same as line 1	4.2e	1*	Δ	4.2e	1*	Δ
6	Dynamic burn-in (method 1015)	4.2c 4.5.2	Req'd			Not req'd	
7	Same as line 1	4.2e	1	Δ			
8	Final electrical parameters		1*, 2, 3, 7, 8, 9			1*, 2, 7, 9 4/	
9	Group A test requirements	4.4.1	1, 2, 3, 4, 7, 9, 10, 11		4.4.1	1, 2, 3, 4, 7, 8, 9, 10, 11	
10	Group B test when using method 5005 QCI option	4.4.2	1, 2, 3, 7, 8, 9, 10, 11	Δ			
11	Group C end-point electrical parameters	4.4.2	1, 2, 3, 7, 8, 9, 10, 11	Δ	4.4.3	1, 2	Δ
12	Group D end-point electrical parameters	4.4.4	1, 2, 3		4.4.4	1, 2	

1/ Blank spaces indicate tests are not applicable.

2/ * indicates PDA applies to subgroup 1 (see 4.2.1).

3/ Δ indicates delta limits shall be required only on table III subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may, at their option, either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup 1 electrical measurement without delta measurements within 24 hours after burn-in completion (removal of bias).

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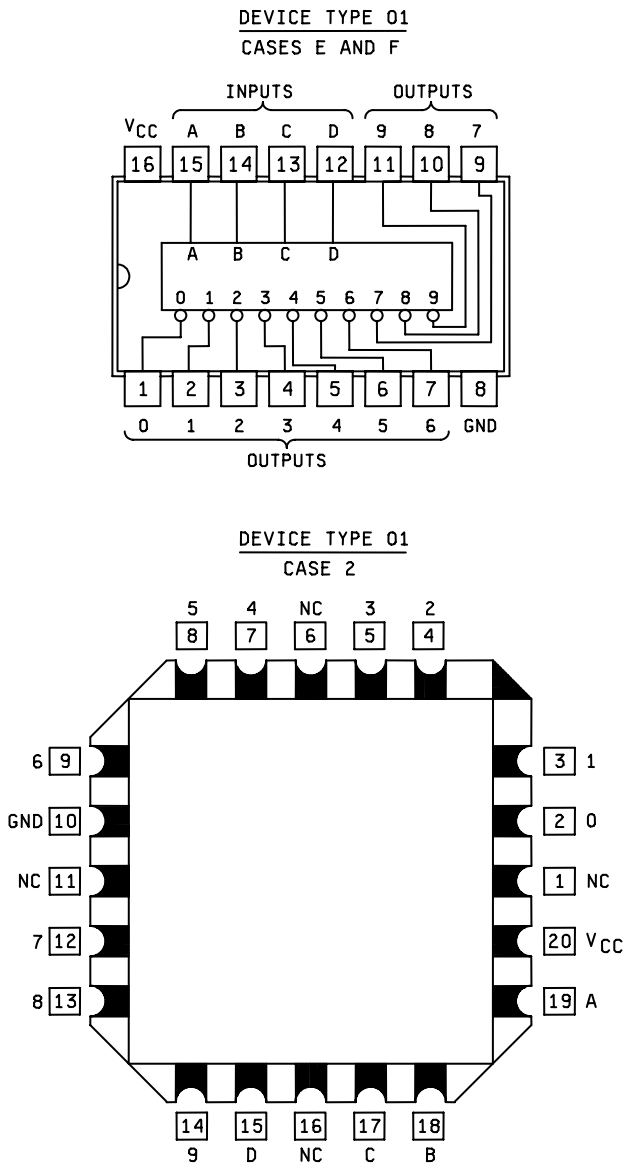
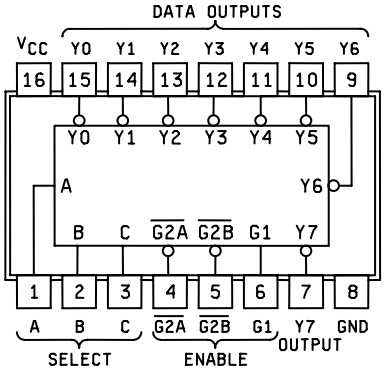


FIGURE 1. Terminal connections (top view).

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DEVICE TYPES 02 AND 52
CASES E AND F



DEVICE TYPES 02 AND 52
CASE 2

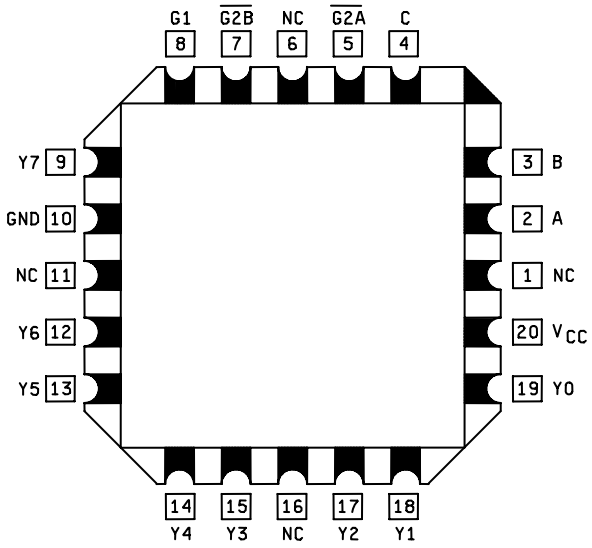
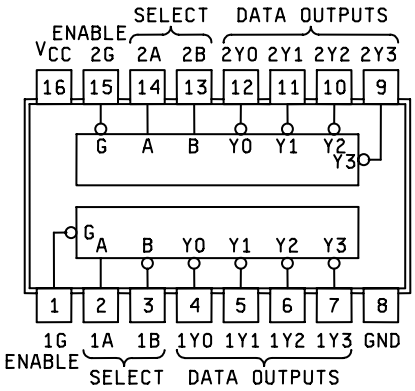


FIGURE 1. Terminal connections (top view) - Continued.

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DEVICE TYPES 03 AND 53
CASES E AND F



DEVICE TYPES 03 AND 53
CASE 2

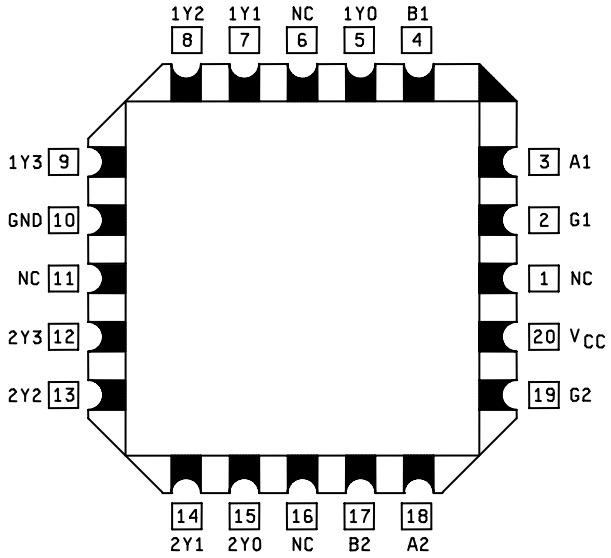
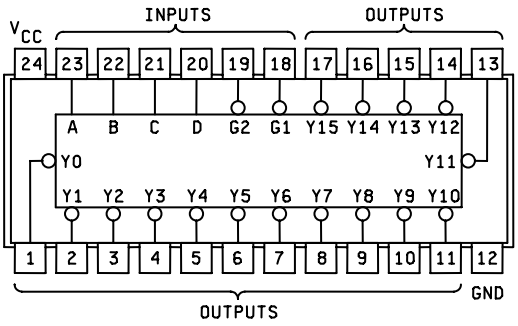


FIGURE 1. Terminal connections (top view) - Continued.

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DEVICE TYPE 04
CASES K AND L



DEVICE TYPE 04
CASE 3

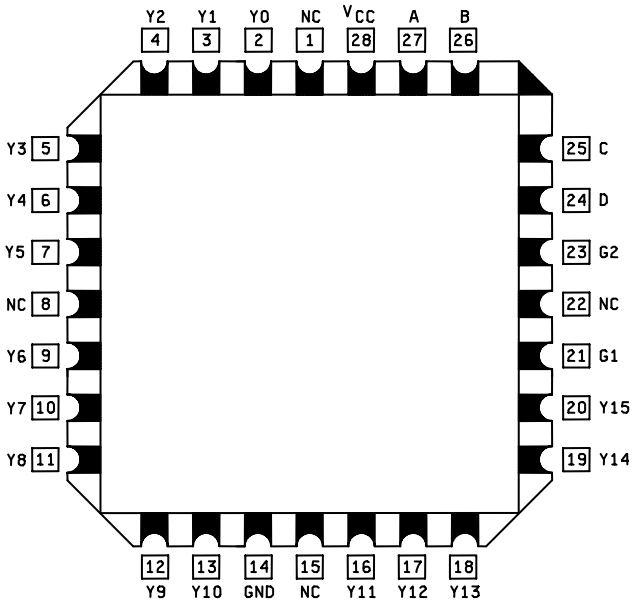


FIGURE 1. Terminal connections (top view) - Continued.

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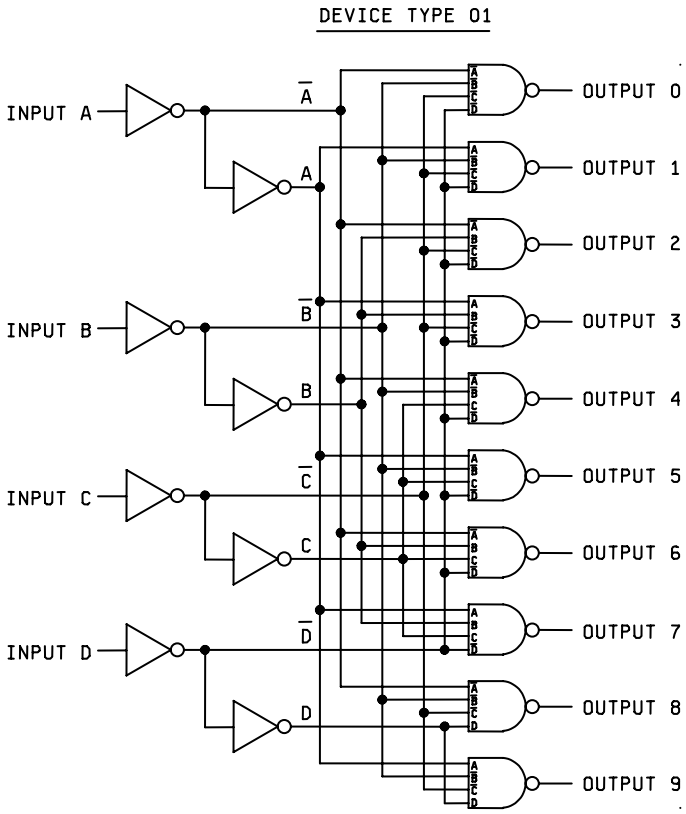


FIGURE 2. Logic diagrams.

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DEVICE TYPES 02 AND 52

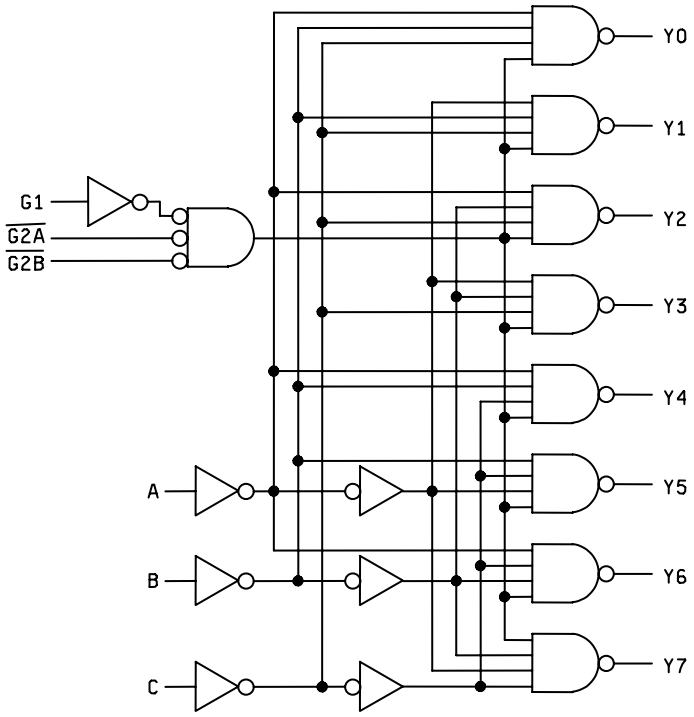


FIGURE 2. Logic diagrams – Continued.

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DEVICE TYPES 03 AND 53

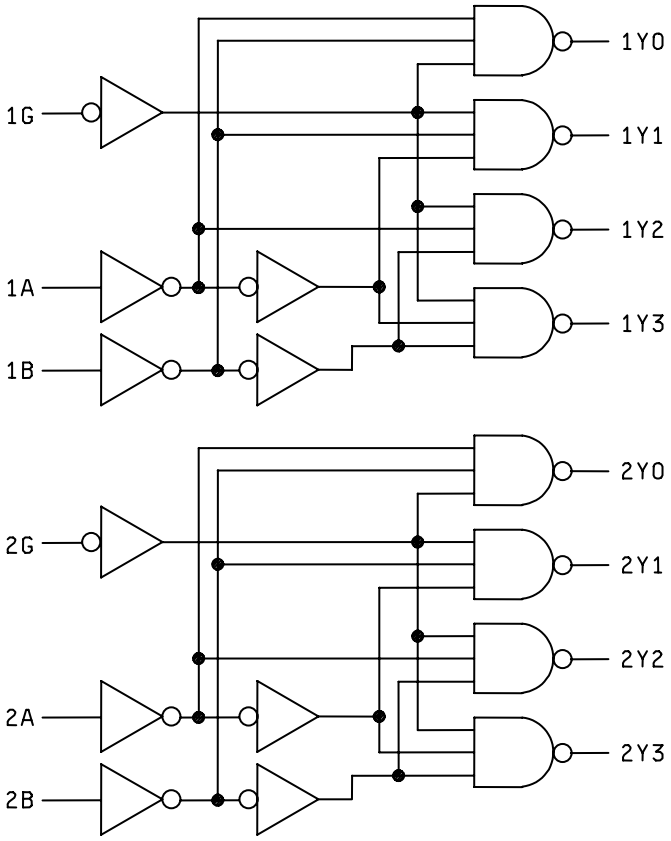


FIGURE 2. Logic diagrams – Continued.

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DEVICE TYPE 04

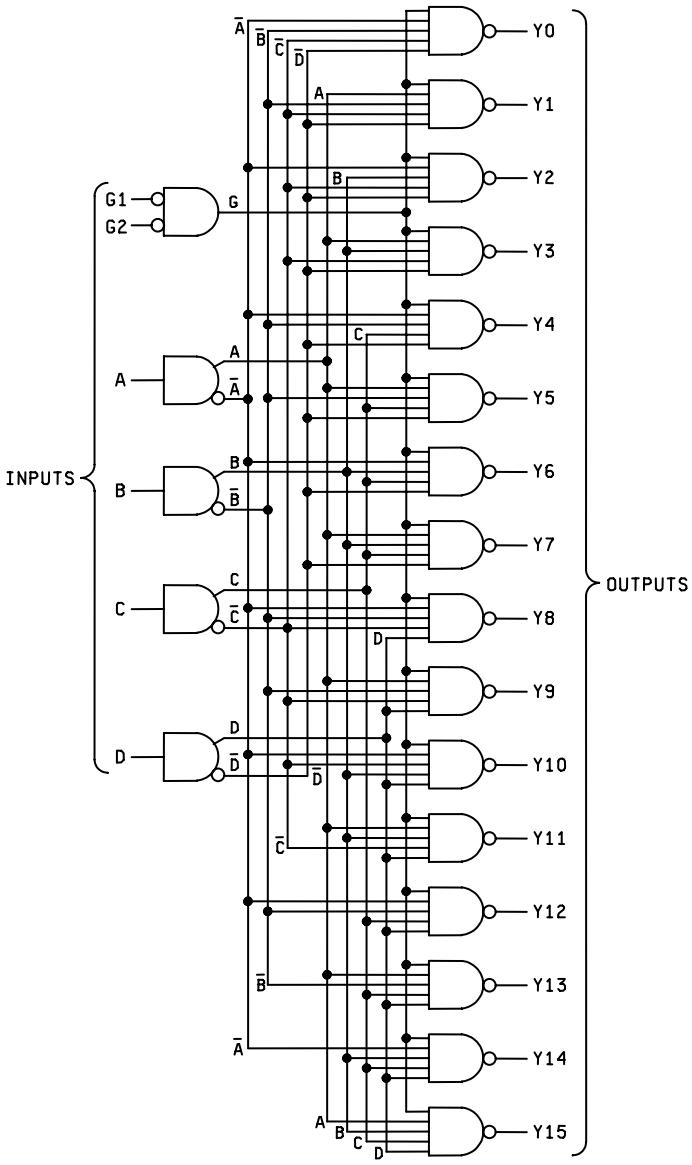


FIGURE 2. Logic diagrams – Continued.

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Device type 01

Inputs				Outputs									
D	C	B	A	0	1	2	3	4	5	6	7	8	9
L	L	L	L	L	H	H	H	H	H	H	H	H	H
L	L	L	H	H	L	H	H	H	H	H	H	H	H
L	L	H	L	H	H	L	H	H	H	H	H	H	H
L	L	H	H	H	H	H	L	H	H	H	H	H	H
L	H	L	L	H	H	H	H	L	H	H	H	H	H
L	H	L	H	H	H	H	H	H	L	H	H	H	H
L	H	H	L	H	H	H	H	H	H	L	H	H	H
L	H	H	H	H	H	H	H	H	H	H	L	H	H
H	L	L	L	H	H	H	H	H	H	H	H	L	H
H	L	L	H	H	H	H	H	H	H	H	H	H	L
H	L	H	L	H	H	H	H	H	H	H	H	H	H
H	L	H	H	H	H	H	H	H	H	H	H	H	H
H	H	L	L	H	H	H	H	H	H	H	H	H	H
H	H	L	H	H	H	H	H	H	H	H	H	H	H
H	H	H	L	H	H	H	H	H	H	H	H	H	H
H	H	H	H	H	H	H	H	H	H	H	H	H	H

H = High level voltage
L = Low level voltage

Device types 02 and 52

Inputs					Outputs							
Enable		Select										
G1	$\overline{G2}$ *	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H
H	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	H	H	H	H	H	L	H	H	H	H
H	L	H	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	H	L	H	H
H	L	H	H	L	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	L

* $\overline{G2} = \overline{G2A} + \overline{G2B}$
H = High level voltage
L = Low level voltage
X = Irrelevant

FIGURE 3. Truth tables.

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Device types 03 and 53

Inputs			Outputs			
Enable	Select					
G	B	A	Y0	Y1	Y2	Y3
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	L	H	H	L	H	H
L	H	L	H	H	L	H
L	H	H	H	H	H	L

H = High level voltage
L = Low level voltage
X = Irrelevant

Device type 04

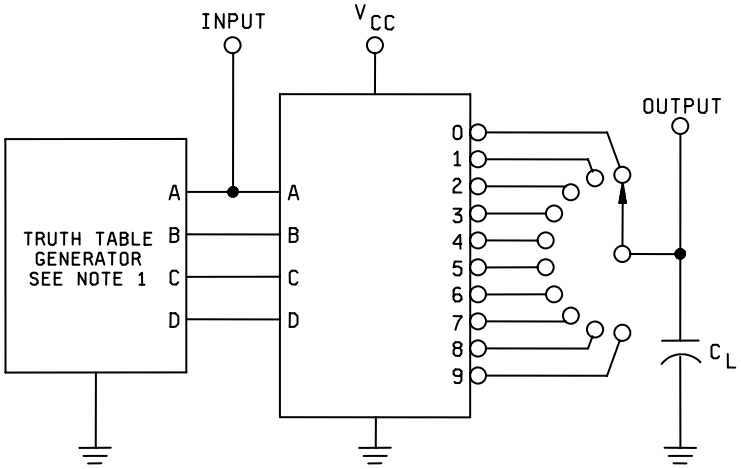
Inputs						Outputs														
G1	G2	D	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11	Y12	Y13	Y14	Y15
L	L	L	L	L	L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	H	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	L	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H
L	L	L	H	L	L	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H
L	L	L	H	L	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H	H
L	L	L	H	H	L	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H	H
L	L	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H	H
L	L	H	L	L	L	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H	H
L	L	H	L	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H	H
L	L	H	L	H	L	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H	H
L	L	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H	H
L	L	H	H	L	L	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H	H
L	L	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H	H
L	L	H	H	H	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L	H
L	L	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	L
L	H	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
H	L	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H
H	H	X	X	X	X	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H

H = High level voltage
L = Low level voltage
X = Irrelevant

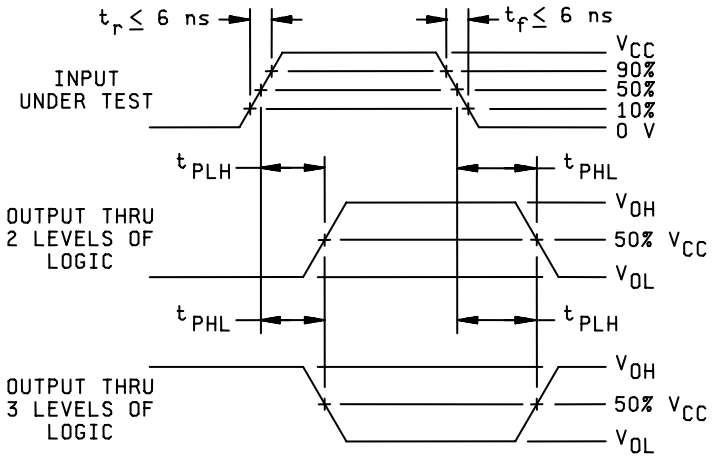
FIGURE 3. Truth tables – Continued.

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DEVICE TYPE 01



TEST CIRCUIT



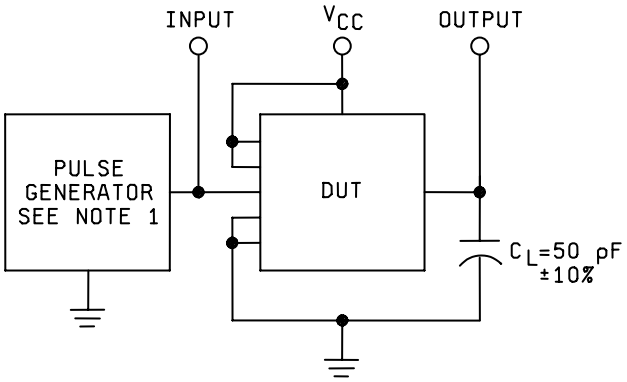
NOTES:

1. The truth table generator has the following characteristics: PRR < 1 MHz, 50% duty cycle. Transition of the input under test shall occur simultaneously with or following the transition of the other inputs, $t_r = t_f \leq 6$ ns.
2. $C_L = 50$ pF $\pm$ 10%; includes probe and jig capacitance.

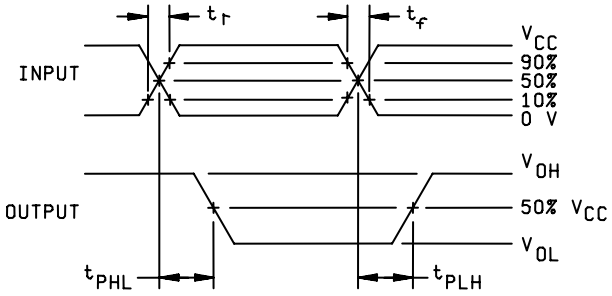
FIGURE 4. Switching time test circuits and waveforms.

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DEVICE TYPES 02,03,52 AND 53



TEST CIRCUIT



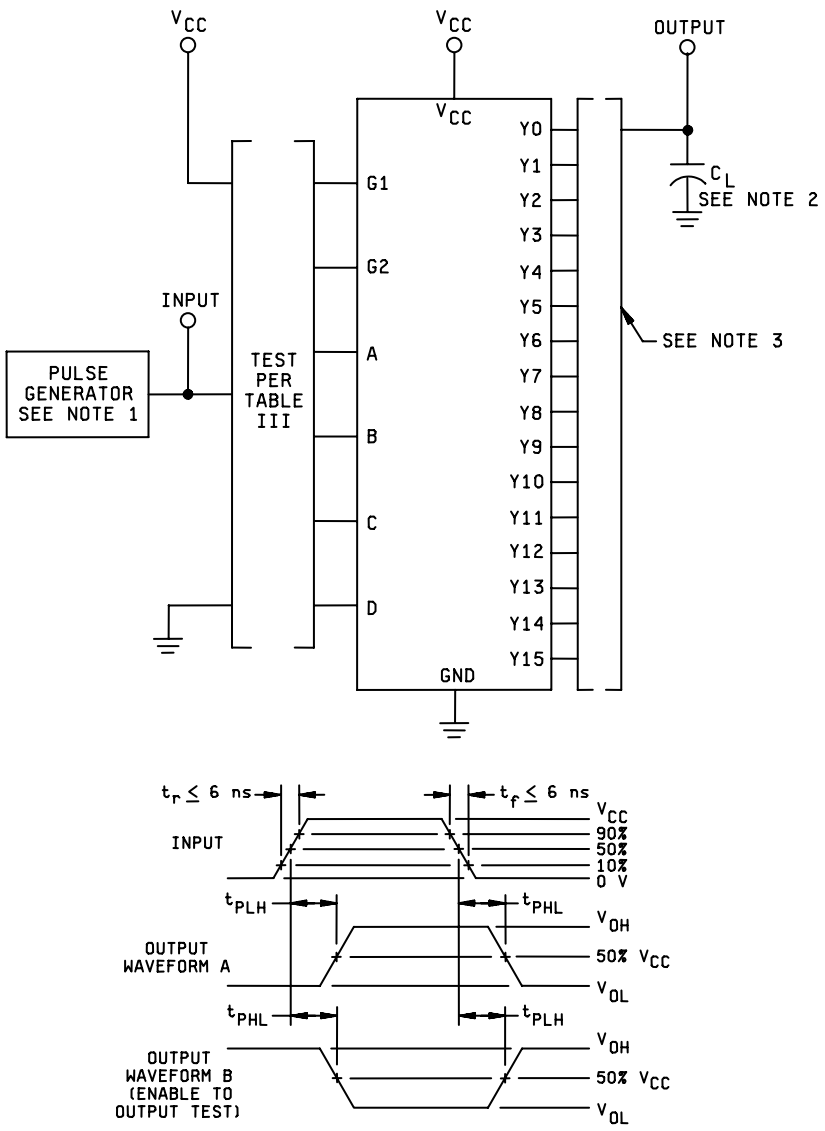
NOTES:

1. Pulse generator has the following characteristics: $t_r = t_f \leq 6$ ns; PRR ≤ 1 MHz; duty cycle = 50%.
2. C_L = 50 pF ± 10%; includes scope probe, wiring, and stray capacitance, without package in test fixture.
3. Voltage measurements are to be made with respect to network ground terminal.
4. All unused inputs are tied to V_{CC} or ground according to table III.
5. The input signal for HCT device types (52 and 53) will be 0 to 3 volts; however, the 50% V_{CC} measuring point is 1.3 volts for input and output.

FIGURE 4. Switching time test circuits and waveforms – Continued.

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DEVICE TYPE 04



NOTES:

1. Pulse generator has the following characteristics: $t_r = t_f \leq 6$ ns; PRR ≤ 1 MHz; duty cycle = 50%.
2. C_L = 50 pF ± 10%; includes probe and jig capacitance.
3. Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.

FIGURE 4. Switching time test circuits and waveforms – Continued.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1								1/				1mA				GND	D		1.5					V
		2								"					1mA			"	C		"					"
		3								"						1mA		"	B		"					"
		4								"							1mA	"	A		"					"
V_{IC}(neg)		5								GND				-1mA				1/	D		-1.5					"
		6								"					-1mA			"	C		"					"
		7								"						-1mA		"	B		"					"
		8								"							-1mA	"	A		"					"
I_{CC}	3005	9								"				6.0V	6.0V	6.0V	6.0V	6.0V	V_{CC}		0.2		20			µA
V_{OH3}	3006	10	-20µA							"				4.2V	4.2V	4.2V	4.2V	"	0	5.95		5.95		5.95		V
		11		-20µA						"				"	"	"	"	"	1	"		"		"		"
		12			-20µA					"				"	"	"	"	"	2	"		"		"		"
		13				-20µA				"				"	"	"	"	"	3	"		"		"		"
		14					-20µA			"				"	"	"	"	"	4	"		"		"		"
		15						-20µA		"				"	"	"	"	"	5	"		"		"		"
		16							-20µA	"				"	"	"	"	"	6	"		"		"		"
		17								"	-20µA			"	"	"	"	"	7	"		"		"		"
		18								"		-20µA		"	"	"	"	"	8	"		"		"		"
		19								"			-20µA	"	"	"	"	"	9	"		"		"		"
V_{OH5}	3006	20	-5.2mA							"				"	"	"	"	"	0	5.48		5.2		5.48		"
		21		-5.2mA						"				"	"	"	"	"	1	"		"		"		"
		22			-5.2mA					"				"	"	"	"	"	2	"		"		"		"
		23				-5.2mA				"				"	"	"	"	"	3	"		"		"		"
		24					-5.2mA			"				"	"	"	"	"	4	"		"		"		"
		25						-5.2mA		"				"	"	"	"	"	5	"		"		"		"
		26							-5.2mA	"				"	"	"	"	"	6	"		"		"		"
		27								"	-5.2mA			"	"	"	"	"	7	"		"		"		"
		28								"		-5.2mA		"	"	"	"	"	8	"		"		"		"
		29								"			-5.2mA	"	"	"	"	"	9	"		"		"		"
V_{OL3}	3007	30	20µA							"				1.2V	1.2V	1.2V	1.2V	"	0		0.05		0.05		0.05	"
		31		20µA						"				"	"	1.2V	4.2V	"	1		"		"		"	"
		32			20µA					"				"	"	4.2V	1.2V	"	2		"		"		"	"
		33				20µA				"				"	"	4.2V	4.2V	"	3		"		"		"	"
		34					20µA			"				"	4.2V	1.2V	1.2V	"	4		"		"		"	"
		35						20µA		"				"	"	1.2V	4.2V	"	5		"		"		"	"
		36							20µA	"				"	"	4.2V	1.2V	"	6		"		"		"	"
		37								"	20µA			"	"	4.2V	4.2V	"	7		"		"		"	"
		38								"		20µA		4.2V	1.2V	1.2V	1.2V	"	8		"		"		"	"
		39								"			20µA	4.2V	1.2V	1.2V	4.2V	"	9		"		"		"	"
V_{OL5}	3007	40	5.2mA							"				1.2V	1.2V	1.2V	1.2V	"	0		0.26		0.4		0.26	"
		41		5.2mA						"				"	"	1.2V	4.2V	"	1		"		"		"	"
		42			5.2mA					"				"	"	4.2V	1.2V	"	2		"		"		"	"
		43				5.2mA				"				"	"	4.2V	4.2V	"	3		"		"		"	"
		44					5.2mA			"				"	4.2V	1.2V	1.2V	"	4		"		"		"	"
		45						5.2mA		"				"	"	1.2V	4.2V	"	5		"		"		"	"
		46							5.2mA	"				"	"	4.2V	1.2V	"	6		"		"		"	"
		47								"	5.2mA			"	"	4.2V	4.2V	"	7		"		"		"	"
		48								"		5.2mA		4.2V	1.2V	1.2V	1.2V	"	8		"		"		"	"
		49								"			5.2mA	4.2V	1.2V	1.2V	4.2V	"	9		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	50	GND							GND				4.0V	4.0V	4.0V	4.0V	4.0V	0	-10	-120	-10	-120	-10	-120	mA
		51		GND						"				"	"	"	"	"	1	"	"	"	"	"	"	"
		52			GND					"				"	"	"	"	"	2	"	"	"	"	"	"	"
		53				GND				"				"	"	"	"	"	3	"	"	"	"	"	"	"
		54					GND			"				"	"	"	"	"	4	"	"	"	"	"	"	"
		55						GND		"				"	"	"	"	"	5	"	"	"	"	"	"	"
		56							GND	"				"	"	"	"	"	6	"	"	"	"	"	"	"
		57								"	GND			"	"	"	"	"	7	"	"	"	"	"	"	"
		58								"		GND		"	"	"	"	"	8	"	"	"	"	"	"	"
		59								"			GND	"	"	"	"	"	9	"	"	"	"	"	"	"
I_{IH}	3010	60								"				6.0V	GND	GND	GND	6.0V	D		.05		.01			µA
		61								"				GND	6.0V	GND	"	"	C		"		"			"
		62								"				"	GND	6.0V	"	"	B		"		"			"
		63								"				"	GND	GND	6.0V	"	A		"		"			"
I_{IL}	3009	64								"				6.0V	6.0V	6.0V	GND	"	A		-.05		-.01			"
		65								"				"	6.0V	GND	6.0V	"	B		"		"			"
		66								"				"	GND	6.0V	"	"	C		"		"			"
		67								"				GND	6.0V	6.0V	"	"	D		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_i	3012	68								GND				2/	2/			GND	D		10					pF
		69								"						2/		"	C		"					"
		70								"							2/	"	B		"					"
		71								"								"	A		"					"
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table tests 3/	3009	72	L	H	H	H	H	H	H	GND	H	H	H	B	B	B	B	4.5V	4/							
		73	H	L	H	H	"	"	"	"	"	"	"	"	"	B	A	"								
		74	"	H	L	"	"	"	"	"	"	"	"	"	"	A	B	"								
		75	"	"	H	L	"	"	"	"	"	"	"	"	"	"	A	"								
		76	"	"	"	H	L	"	"	"	"	"	"	"	"	"	B	"								
		77	"	"	"	"	H	L	"	"	"	"	"	"	"	"	A	"								
		78	"	"	"	"	"	H	L	"	"	"	"	"	"	"	B	"								
		79	"	"	"	"	"	"	H	"	L	"	"	"	"	"	A	"								
		80	"	"	"	"	"	"	"	"	H	L	"	A	"	"	B	"								
		81	"	"	"	"	"	"	"	"	"	H	L	"	"	"	A	"								
		82	"	"	"	"	"	"	"	"	"	"	H	"	"	"	B	"								
		83	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"								
		84	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"								
		85	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"								
		86	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"								
		87	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"								

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Case 2 / E and F	Terminal conditions 1/ 2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits Subgroup 9 T_C = +25°C 5/ Min	Subgroup 9 Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C 5/ Min	Subgroup 11 Max	Unit
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16									
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}								
t_{PHL1}	3003 (Fig. 4)	88	OUT							GND				GND	GND	GND	IN	4.5V	A to 0	4	26	4	35	4	26	ns
		89	"							"				"	GND	IN	GND	"	B to 0	"	"	"	"	"	"	"
		90	"							"				"	IN	GND	"	"	C to 0	"	"	"	"	"	"	"
		91	"							"				IN	GND	"	"	"	D to 0	"	"	"	"	"	"	"
		92		OUT						"				GND	"	"	IN	"	A to 1	"	"	"	"	"	"	"
		93		"						"				"	"	IN	4.5V	"	B to 1	"	"	"	"	"	"	"
		94		"						"				"	IN	GND	"	"	C to 1	"	"	"	"	"	"	"
		95		"						"				IN	GND	GND	"	"	D to 1	"	"	"	"	"	"	"
		96			OUT					"				GND	"	4.5V	IN	"	A to 2	"	"	"	"	"	"	"
		97			"					"				"	"	IN	GND	"	B to 2	"	"	"	"	"	"	"
		98			"					"				"	IN	4.5V	"	"	C to 2	"	"	"	"	"	"	"
		99								"				IN	GND	"	"	"	D to 2	"	"	"	"	"	"	"
		100				OUT				"				GND	"	"	IN	"	A to 3	"	"	"	"	"	"	"
		101				"				"				"	"	IN	4.5V	"	B to 3	"	"	"	"	"	"	"
		102				"				"				"	IN	4.5V	"	"	C to 3	"	"	"	"	"	"	"
		103								"				IN	GND	4.5V	"	"	D to 3	"	"	"	"	"	"	"
		104					OUT			"				GND	4.5V	GND	IN	"	A to 4	"	"	"	"	"	"	"
		105					"			"				"	4.5V	IN	GND	"	B to 4	"	"	"	"	"	"	"
		106					"			"				"	IN	GND	"	"	C to 4	"	"	"	"	"	"	"
		107								"				IN	4.5V	"	"	"	D to 4	"	"	"	"	"	"	"
		108						OUT		"				GND	"	"	IN	"	A to 5	"	"	"	"	"	"	"
		109						"		"				"	"	IN	4.5V	"	B to 5	"	"	"	"	"	"	"
		110						"		"				"	IN	GND	"	"	C to 5	"	"	"	"	"	"	"
		111						"		"				IN	4.5V	GND	"	"	D to 5	"	"	"	"	"	"	"
		112							OUT	"				GND	"	4.5V	IN	"	A to 6	"	"	"	"	"	"	"
		113							"	"				"	"	IN	GND	"	B to 6	"	"	"	"	"	"	"
		114							"	"				"	IN	4.5V	"	"	C to 6	"	"	"	"	"	"	"
		115							"	"				IN	4.5V	"	"	"	D to 6	"	"	"	"	"	"	"
		116							"	"	OUT			GND	"	"	IN	"	A to 7	"	"	"	"	"	"	"
		117							"	"	"			"	"	IN	4.5V	"	B to 7	"	"	"	"	"	"	"
		118							"	"	"			"	IN	4.5V	"	"	C to 7	"	"	"	"	"	"	"
		119							"	"	"			IN	4.5V	4.5V	"	"	D to 7	"	"	"	"	"	"	"
		120							"	"		OUT		4.5V	GND	GND	IN	"	A to 8	"	"	"	"	"	"	"
		121							"	"		"		"	GND	IN	GND	"	B to 8	"	"	"	"	"	"	"
		122							"	"		"		"	IN	GND	"	"	C to 8	"	"	"	"	"	"	"
		123							"	"		"		IN	GND	"	"	"	D to 8	"	"	"	"	"	"	"
		124							"	"			OUT	4.5V	"	"	IN	"	A to 9	"	"	"	"	"	"	"
		125							"	"			"	"	"	IN	4.5V	"	B to 9	"	"	"	"	"	"	"
		126							"	"			"	"	IN	GND	"	"	C to 9	"	"	"	"	"	"	"
		127							"	"			"	IN	GND	"	"	"	D to 9	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLH1}	3003 (Fig. 4)	128	OUT							GND				GND	GND	GND	IN	4.5V	A to 0	4	26	4	35	4	26	ns
		129	"							"				"	GND	IN	GND	"	B to 0	"	"	"	"	"	"	"
		130	"							"				"	IN	GND	"	"	C to 0	"	"	"	"	"	"	"
		131	"							"				IN	GND	"	"	"	D to 0	"	"	"	"	"	"	"
		132		OUT						"				GND	"	"	IN	"	A to 1	"	"	"	"	"	"	"
		133		"						"				"	"	IN	4.5V	"	B to 1	"	"	"	"	"	"	"
		134		"						"				"	IN	GND	"	"	C to 1	"	"	"	"	"	"	"
		135		"						"				IN	GND	GND	"	"	D to 1	"	"	"	"	"	"	"
		136			OUT					"				GND	"	4.5V	IN	"	A to 2	"	"	"	"	"	"	"
		137			"					"				"	"	IN	GND	"	B to 2	"	"	"	"	"	"	"
		138			"					"				"	IN	4.5V	"	"	C to 2	"	"	"	"	"	"	"
		139			"					"				IN	GND	"	"	"	D to 2	"	"	"	"	"	"	"
		140				OUT				"				GND	"	"	IN	"	A to 3	"	"	"	"	"	"	"
		141				"				"				"	"	IN	4.5V	"	B to 3	"	"	"	"	"	"	"
		142				"				"				"	IN	4.5V	"	"	C to 3	"	"	"	"	"	"	"
		143				"				"				IN	GND	4.5V	"	"	D to 3	"	"	"	"	"	"	"
		144					OUT			"				GND	4.5V	GND	IN	"	A to 4	"	"	"	"	"	"	"
		145					"			"				"	4.5V	IN	GND	"	B to 4	"	"	"	"	"	"	"
		146					"			"				"	IN	GND	"	"	C to 4	"	"	"	"	"	"	"
		147					"			"				IN	4.5V	"	"	"	D to 4	"	"	"	"	"	"	"
		148						OUT		"				GND	"	"	IN	"	A to 5	"	"	"	"	"	"	"
		149						"		"				"	"	IN	4.5V	"	B to 5	"	"	"	"	"	"	"
		150						"		"				"	IN	GND	"	"	C to 5	"	"	"	"	"	"	"
		151						"		"				IN	4.5V	GND	"	"	D to 5	"	"	"	"	"	"	"
		152							OUT	"				GND	"	4.5V	IN	"	A to 6	"	"	"	"	"	"	"
		153							"	"				"	"	IN	GND	"	B to 6	"	"	"	"	"	"	"
		154							"	"				"	IN	4.5V	"	"	C to 6	"	"	"	"	"	"	"
		155							"	"				IN	4.5V	"	"	"	D to 6	"	"	"	"	"	"	"
		156								"	OUT			GND	"	"	IN	"	A to 7	"	"	"	"	"	"	"
		157								"	"			"	"	IN	4.5V	"	B to 7	"	"	"	"	"	"	"
		158								"	"			"	IN	4.5V	"	"	C to 7	"	"	"	"	"	"	"
		159								"	"			IN	4.5V	4.5V	"	"	D to 7	"	"	"	"	"	"	"
		160								"		OUT		4.5V	GND	GND	IN	"	A to 8	"	"	"	"	"	"	"
		161								"		"		"	GND	IN	GND	"	B to 8	"	"	"	"	"	"	"
		162								"		"		"	IN	GND	"	"	C to 8	"	"	"	"	"	"	"
		163								"		"		IN	GND	"	"	"	D to 8	"	"	"	"	"	"	"
		164								"		"	OUT	4.5V	"	"	IN	"	A to 9	"	"	"	"	"	"	"
		165								"		"	"	"	"	IN	4.5V	"	B to 9	"	"	"	"	"	"	"
		166								"		"	"	"	IN	GND	"	"	C to 9	"	"	"	"	"	"	"
		167								"		"	"	IN	GND	"	"	"	D to 9	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 4)	168	OUT							"				GND	"	"	IN	"	0	3	15	3	20	3	15	"
		169		OUT						"				"	"	"	"	"	1	"	"	"	"	"	"	"
		170			OUT					"				"	"	4.5V	"	"	2	"	"	"	"	"	"	"
		171				OUT				"				"	"	4.5V	"	"	3	"	"	"	"	"	"	"
		172					OUT			"				"	4.5V	GND	"	"	4	"	"	"	"	"	"	"
		173						OUT		"				"	"	GND	"	"	5	"	"	"	"	"	"	"
		174							OUT	"				"	"	4.5V	"	"	6	"	"	"	"	"	"	"
		175								"	OUT			"	"	4.5V	"	"	7	"	"	"	"	"	"	"
		176								"		OUT		4.5V	GND	GND	"	"	8	"	"	"	"	"	"	"
		177								"			OUT	4.5V	"	"	"	"	9	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/															Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	0	1	2	3	4	5	6	GND	7	8	9	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 4)	178	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	GND	GND	GND	IN	4.5V	0	3	15	3	20	3	15	ns
		179								"				"	"	"	"	"	1	"	"	"	"	"	"	"
		180								"				"	"	4.5V	"	"	2	"	"	"	"	"	"	"
		181								"				"	"	4.5V	"	"	3	"	"	"	"	"	"	"
		182								"				"	4.5V	GND	"	"	4	"	"	"	"	"	"	"
		183								"				"	"	GND	"	"	5	"	"	"	"	"	"	"
		184								"				"	"	4.5V	"	"	6	"	"	"	"	"	"	"
		185								"				"	"	4.5V	"	"	7	"	"	"	"	"	"	"
		186								"				4.5V	GND	GND	"	"	8	"	"	"	"	"	"	"
		187								"				4.5V	"	"	"	"	9	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA							1/									A		1.5					V
		2		1mA						"									B		"					"
		3			1mA					"									C		"					"
		4				1mA				"									$\overline{G2A}$		"					"
		5					1mA			"									G2B		"					"
		6						1mA		"									G1		"					"
V_{IC}(neg)		7	-1mA							GND								1/	A		-1.5					"
		8		-1mA						"								"	B		"					"
		9			-1mA					"								"	C		"					"
		10				-1mA				"								"	$\overline{G2A}$		"					"
		11					-1mA			"								"	G2B		"					"
		12						-1mA		"								"	G1		"					"
I_{CC}	3005	13	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V		"								6.0V	V_{CC}		0.2		20			µA
V_{OH3}	3006	14	1.2V	1.2V	1.2V	4.2V	1.2V	1.2V		"							-20µA	"	Y0	5.95		5.95		5.95		V
		15	"	"	"	"	"	"		"						-20µA		"	Y1	"		"		"		"
		16	"	"	"	"	"	"		"					-20µA			"	Y2	"		"		"		"
		17	"	"	"	"	"	"		"				-20µA				"	Y3	"		"		"		"
		18	"	"	"	"	"	"		"			-20µA					"	Y4	"		"		"		"
		19	"	"	"	"	"	"		"		-20µA						"	Y5	"		"		"		"
		20	"	"	"	"	"	"		"	-20µA							"	Y6	"		"		"		"
		21	"	"	"	"	"	"	-20µA	"								"	Y7	"		"		"		"
V_{OH5}	3006	22	"	"	"	"	"	"		"							-5.2mA	"	Y0	5.48		5.2		5.48		"
		23	"	"	"	"	"	"		"						-5.2mA		"	Y1	"		"		"		"
		24	"	"	"	"	"	"		"					-5.2mA			"	Y2	"		"		"		"
		25	"	"	"	"	"	"		"				-5.2mA				"	Y3	"		"		"		"
		26	"	"	"	"	"	"		"			-5.2mA					"	Y4	"		"		"		"
		27	"	"	"	"	"	"		"		-5.2mA						"	Y5	"		"		"		"
		28	"	"	"	"	"	"		"	-5.2mA							"	Y6	"		"		"		"
		29	"	"	"	"	"	"	-5.2mA	"								"	Y7	"		"		"		"
V_{OL3}	3007	30	"	"	"	1.2V	"	4.2V		"						20µA		"	Y0		0.05		0.05		0.05	"
		31	4.2V	"	"	"	"	"		"					20µA			"	Y1		"		"		"	"
		32	1.2V	4.2V	"	"	"	"		"				20µA				"	Y2		"		"		"	"
		33	4.2V	4.2V	"	"	"	"		"			20µA					"	Y3		"		"		"	"
		34	1.2V	1.2V	4.2V	"	"	"		"		20µA						"	Y4		"		"		"	"
		35	4.2V	1.2V	"	"	"	"		"	20µA							"	Y5		"		"		"	"
		36	1.2V	4.2V	"	"	"	"		"								"	Y6		"		"		"	"
		37	4.2V	4.2V	"	"	"	"	20µA	"								"	Y7		"		"		"	"
V_{OL5}	3007	38	1.2V	1.2V	1.2V	"	"	"		"						5.2mA		"	Y0		0.26		0.4		0.26	"
		39	4.2V	1.2V	"	"	"	"		"					5.2mA			"	Y1		"		"		"	"
		40	1.2V	4.2V	"	"	"	"		"				5.2mA				"	Y2		"		"		"	"
		41	4.2V	4.2V	"	"	"	"		"			5.2mA					"	Y3		"		"		"	"
		42	1.2V	1.2V	4.2V	"	"	"		"		5.2mA						"	Y4		"		"		"	"
		43	4.2V	1.2V	"	"	"	"		"	5.2mA							"	Y5		"		"		"	"
		44	1.2V	4.2V	"	"	"	"		"								"	Y6		"		"		"	"
		45	4.2V	4.2V	"	"	"	"	5.2mA	"								"	Y7		"		"		"	"

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Case 2	Terminal conditions 1/																Measured terminal	Test limits						Unit
		E and F	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	Min	Max	Min	Max	
			A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}								
I_{OS4}	3011	46	GND	GND	GND	4.0V	4.0V	GND		GND								4.0V	Y0	-10	-120	-10	-120	-10	-120	mA
		47	"	"	"	"	"	"		"								"	Y1	"	"	"	"	"	"	"
		48	"	"	"	"	"	"		"								"	Y2	"	"	"	"	"	"	"
		49	"	"	"	"	"	"		"								"	Y3	"	"	"	"	"	"	"
		50	"	"	"	"	"	"		"								"	Y4	"	"	"	"	"	"	"
		51	"	"	"	"	"	"		"								"	Y5	"	"	"	"	"	"	"
		52	"	"	"	"	"	"		"								"	Y6	"	"	"	"	"	"	"
		53	"	"	"	"	"	"	GND	"								"	Y7	"	"	"	"	"	"	"
I_{IH}	3010	54	6.0V	GND	GND	GND	GND	GND		"								6.0V	A		0.05		0.1			µA
		55	GND	6.0V	GND	"	"	"		"								"	B		"		"			"
		56	"	GND	6.0V	"	"	"		"								"	C		"		"			"
		57	"	"	GND	6.0V	"	"		"								"	$\overline{G2A}$		"		"			"
		58	"	"	"	GND	6.0V	"		"								"	G2B		"		"			"
		59	"	"	"	GND	GND	6.0V		"								"	G1		"		"			"
I_{IL}	3009	60	"	6.0V	6.0V	6.0V	6.0V	"		"								"	A		-0.05		-0.1			"
		61	6.0V	GND	6.0V	"	"	"		"								"	B		"		"			"
		62	"	6.0V	GND	"	"	"		"								"	C		"		"			"
		63	"	"	6.0V	GND	"	"		"								"	$\overline{G2A}$		"		"			"
		64	"	"	"	6.0V	GND	"		"								"	G2B		"		"			"
		65	"	"	"	6.0V	6.0V	GND		"								"	G1		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_i	3012	66	2/							GND								GND	A		10					pF
		67		2/						"								"	B		"					"
		68			2/					"								"	C		"					"
		69				2/				"								"	$\overline{G2A}$		"					"
		70					2/			"								"	G2B		"					"
		71						2/		"								"	G1		"					"
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table tests 3/	3014	72	A	A	A	A	A	A	A	H	GND	H	H	H	H	H	H	H	4.5V	4/						
		73	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		74	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		75	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		76	"	"	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		77	"	"	"	A	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"						
		78	"	"	"	B	B	A	B	"	"	"	"	"	"	"	"	"	"	"	"						
		79	"	"	"	B	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"						
		80	"	"	"	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"						
		81	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	L	"	"						
		82	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	H	"	"						
		83	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"						
		84	"	"	"	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"	"						
		85	"	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"	"	"	"						
		86	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"	"	"	"	"						
		87	"	"	"	"	"	"	"	"	"	L	H	H	"	"	"	"	"	"	"						
		88	"	"	"	"	"	"	L	"	L	H	"	"	"	"	"	"	"	"	"						

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 4)	89	IN	GND	GND	GND	GND	4.5V		GND							OUT	4.5V	A to Y0	5	35	5	47	5	35	ns
t_{PHL1}		90	GND	IN	GND	"	"	"		"							"	"	B to Y0	"	"	"	"	"	"	"
t_{PHL1}		91	"	GND	IN	"	"	"		"							"	"	C to Y0	"	"	"	"	"	"	"
t_{PHL2}		92	"	"	GND	IN	"	"		"							"	"	G2A to Y0	4	31	4	41	4	31	"
t_{PHL2}		93	"	"	"	GND	IN	"		"							"	"	G2B to Y0	"	"	"	"	"	"	"
t_{PHL2}		94	"	"	"	"	GND	IN		"							"	"	G1 to Y0	"	"	"	"	"	"	"
t_{PHL1}		95	IN	"	"	"	"	4.5V		"						OUT		"	A to Y1	5	35	5	47	5	35	"
t_{PHL1}		96	4.5V	IN	"	"	"	"		"						"		"	B to Y1	"	"	"	"	"	"	"
t_{PHL1}		97	"	GND	IN	"	"	"		"						"		"	C to Y1	"	"	"	"	"	"	"
t_{PHL2}		98	"	"	GND	IN	"	"		"						"		"	G2A to Y1	4	31	4	41	4	31	"
t_{PHL2}		99	"	"	"	GND	IN	"		"						"		"	G2B to Y1	"	"	"	"	"	"	"
t_{PHL2}		100	"	"	"	"	GND	IN		"						"		"	G1 to Y1	"	"	"	"	"	"	"
t_{PHL1}		101	IN	4.5V	"	"	"	4.5V		"					OUT			"	A to Y2	5	35	5	47	5	35	"
t_{PHL1}		102	GND	IN	"	"	"	"		"					"			"	B to Y2	"	"	"	"	"	"	"
t_{PHL1}		103	"	4.5V	IN	"	"	"		"					"			"	C to Y2	"	"	"	"	"	"	"
t_{PHL2}		104	"	"	GND	IN	"	"		"					"			"	G2A to Y2	4	31	4	41	4	31	"
t_{PHL2}		105	"	"	"	GND	IN	"		"					"			"	G2B to Y2	"	"	"	"	"	"	"
t_{PHL2}		106	"	"	"	"	GND	IN		"					"			"	G1 to Y2	"	"	"	"	"	"	"
t_{PHL1}		107	IN	"	"	"	"	4.5V		"				OUT				"	A to Y3	5	35	5	47	5	35	"
t_{PHL1}		108	4.5V	IN	"	"	"	"		"				"				"	B to Y3	"	"	"	"	"	"	"
t_{PHL1}		109	"	4.5V	IN	"	"	"		"				"				"	C to Y3	"	"	"	"	"	"	"
t_{PHL2}		110	"	"	"	IN	"	"		"				"				"	G2A to Y3	4	31	4	41	4	31	"
t_{PHL2}		111	"	"	"	GND	IN	"		"				"				"	G2B to Y3	"	"	"	"	"	"	"
t_{PHL2}		112	"	"	"	"	GND	IN		"				"				"	G1 to Y3	"	"	"	"	"	"	"
t_{PHL1}		113	IN	GND	4.5V	"	"	4.5V		"			OUT					"	A to Y4	5	35	5	47	5	35	"
t_{PHL1}		114	GND	IN	4.5V	"	"	"		"			"					"	B to Y4	"	"	"	"	"	"	"
t_{PHL1}		115	"	GND	IN	"	"	"		"			"					"	C to Y4	"	"	"	"	"	"	"
t_{PHL2}		116	"	"	4.5V	IN	"	"		"			"					"	G2A to Y4	4	31	4	41	4	31	"
t_{PHL2}		117	"	"	"	GND	IN	"		"			"					"	G2B to Y4	"	"	"	"	"	"	"
t_{PHL2}		118	"	"	"	"	GND	IN		"			"					"	G1 to Y4	"	"	"	"	"	"	"
t_{PHL1}		119	IN	GND	"	"	"	4.5V		"		OUT						"	A to Y5	5	35	5	47	5	35	"
t_{PHL1}		120	4.5V	IN	"	"	"	"		"		"						"	B to Y5	"	"	"	"	"	"	"
t_{PHL1}		121	"	GND	IN	"	"	"		"		"						"	C to Y5	"	"	"	"	"	"	"
t_{PHL2}		122	"	"	4.5V	IN	"	"		"		"						"	G2A to Y5	4	31	4	41	4	31	"
t_{PHL2}		123	"	"	"	GND	IN	"		"		"						"	G2B to Y5	"	"	"	"	"	"	"
t_{PHL2}		124	"	"	"	"	GND	IN		"		"						"	G1 to Y5	"	"	"	"	"	"	"
t_{PHL1}		125	IN	4.5V	"	"	"	4.5V		"	OUT							"	A to Y6	5	35	5	47	5	35	"
t_{PHL1}		126	GND	IN	"	"	"	"		"	"							"	B to Y6	"	"	"	"	"	"	"
t_{PHL1}		127	"	4.5V	IN	"	"	"		"	"							"	C to Y6	"	"	"	"	"	"	"
t_{PHL2}		128	"	"	4.5V	IN	"	"		"	"							"	G2A to Y6	4	31	4	41	4	31	"
t_{PHL2}		129	"	"	"	GND	IN	"		"	"							"	G2B to Y6	"	"	"	"	"	"	"
t_{PHL2}		130	"	"	"	"	GND	IN		"	"							"	G1 to Y6	"	"	"	"	"	"	"
t_{PHL1}		131	IN	"	"	"	"	4.5V	OUT	"								"	A to Y7	5	35	5	47	5	35	"
t_{PHL1}		132	4.5V	IN	"	"	"	"	"	"								"	B to Y7	"	"	"	"	"	"	"
t_{PHL1}		133	"	4.5V	IN	"	"	"	"	"								"	C to Y7	"	"	"	"	"	"	"
t_{PHL2}		134	"	"	4.5V	IN	"	"	"	"								"	G2A to Y7	4	31	4	41	4	31	"
t_{PHL2}		135	"	"	"	GND	IN	"	"	"								"	G2B to Y7	"	"	"	"	"	"	"
t_{PHL2}		136	"	"	"	"	GND	IN	"	"								"	G1 to Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Case 2 / E and F / Test no.	Terminal conditions 1/ 2 / 1 / A	3 / 2 / B	4 / 3 / C	5 / 4 / $\overline{G2A}$	7 / 5 / $\overline{G2B}$	8 / 6 / G1	9 / 7 / Y7	10 / 8 / GND	12 / 9 / Y6	13 / 10 / Y5	14 / 11 / Y4	15 / 12 / Y3	17 / 13 / Y2	18 / 14 / Y1	19 / 15 / Y0	20 / 16 / V_{CC}	Measured terminal	Test limits: Subgroup 9 T_C = +25°C 5/ Min	Subgroup 9 Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C 5/ Min	Subgroup 11 Max	Unit
t_{PLH1}	3003 (Fig. 4)	137	IN	GND	GND	GND	GND	4.5V		GND							OUT	4.5V	A to Y0	4	35	4	47	4	35	ns
t_{PLH1}		138	GND	IN	GND	"	"	"		"							"	"	B to Y0	"	"	"	"	"	"	"
t_{PLH1}		139	"	GND	IN	"	"	"		"							"	"	C to Y0	"	"	"	"	"	"	"
t_{PLH2}		140	"	"	GND	IN	"	"		"							"	"	G2A to Y0	"	31	"	41	"	31	"
t_{PLH2}		141	"	"	"	GND	IN	"		"							"	"	G2B to Y0	"	"	"	"	"	"	"
t_{PLH2}		142	"	"	"	"	GND	IN		"							"	"	G1 to Y0	"	"	"	"	"	"	"
t_{PLH1}		143	IN	"	"	"	"	4.5V		"						OUT		"	A to Y1	"	35	"	47	"	35	"
t_{PLH1}		144	4.5V	IN	"	"	"	"		"						"		"	B to Y1	"	"	"	"	"	"	"
t_{PLH1}		145	"	GND	IN	"	"	"		"						"		"	C to Y1	"	"	"	"	"	"	"
t_{PLH2}		146	"	"	GND	IN	"	"		"						"		"	G2A to Y1	"	31	"	41	"	31	"
t_{PLH2}		147	"	"	"	GND	IN	"		"						"		"	G2B to Y1	"	"	"	"	"	"	"
t_{PLH2}		148	"	"	"	"	GND	IN		"						"		"	G1 to Y1	"	"	"	"	"	"	"
t_{PLH1}		149	IN	4.5V	"	"	"	4.5V		"					OUT			"	A to Y2	"	35	"	47	"	35	"
t_{PLH1}		150	GND	IN	"	"	"	"		"					"			"	B to Y2	"	"	"	"	"	"	"
t_{PLH1}		151	"	4.5V	IN	"	"	"		"					"			"	C to Y2	"	"	"	"	"	"	"
t_{PLH2}		152	"	"	GND	IN	"	"		"					"			"	G2A to Y2	"	31	"	41	"	31	"
t_{PLH2}		153	"	"	"	GND	IN	"		"					"			"	G2B to Y2	"	"	"	"	"	"	"
t_{PLH2}		154	"	"	"	"	GND	IN		"					"			"	G1 to Y2	"	"	"	"	"	"	"
t_{PLH1}		155	IN	"	"	"	"	4.5V		"				OUT				"	A to Y3	"	35	"	47	"	35	"
t_{PLH1}		156	4.5V	IN	"	"	"	"		"				"				"	B to Y3	"	"	"	"	"	"	"
t_{PLH1}		157	"	4.5V	IN	"	"	"		"				"				"	C to Y3	"	"	"	"	"	"	"
t_{PLH2}		158	"	"	GND	IN	"	"		"				"				"	G2A to Y3	"	31	"	41	"	31	"
t_{PLH2}		159	"	"	"	GND	IN	"		"				"				"	G2B to Y3	"	"	"	"	"	"	"
t_{PLH2}		160	"	"	"	"	GND	IN		"				"				"	G1 to Y3	"	"	"	"	"	"	"
t_{PLH1}		161	IN	GND	4.5V	"	"	4.5V		"			OUT					"	A to Y4	"	35	"	47	"	35	"
t_{PLH1}		162	GND	IN	4.5V	"	"	"		"			"					"	B to Y4	"	"	"	"	"	"	"
t_{PLH1}		163	"	GND	IN	"	"	"		"			"					"	C to Y4	"	"	"	"	"	"	"
t_{PLH2}		164	"	"	4.5V	IN	"	"		"			"					"	G2A to Y4	"	31	"	41	"	31	"
t_{PLH2}		165	"	"	"	GND	IN	"		"			"					"	G2B to Y4	"	"	"	"	"	"	"
t_{PLH2}		166	"	"	"	"	GND	IN		"			"					"	G1 to Y4	"	"	"	"	"	"	"
t_{PLH1}		167	IN	GND	"	"	"	4.5V		"		OUT						"	A to Y5	"	35	"	47	"	35	"
t_{PLH1}		168	4.5V	IN	"	"	"	"		"		"						"	B to Y5	"	"	"	"	"	"	"
t_{PLH1}		169	"	GND	IN	"	"	"		"		"						"	C to Y5	"	"	"	"	"	"	"
t_{PLH2}		170	"	"	4.5V	IN	"	"		"		"						"	G2A to Y5	"	31	"	41	"	31	"
t_{PLH2}		171	"	"	"	GND	IN	"		"		"						"	G2B to Y5	"	"	"	"	"	"	"
t_{PLH2}		172	"	"	"	"	GND	IN		"		"						"	G1 to Y5	"	"	"	"	"	"	"
t_{PLH1}		173	IN	4.5V	"	"	"	4.5V		"	OUT							"	A to Y6	"	35	"	47	"	35	"
t_{PLH1}		174	GND	IN	"	"	"	"		"	"							"	B to Y6	"	"	"	"	"	"	"
t_{PLH1}		175	"	4.5V	IN	"	"	"		"	"							"	C to Y6	"	"	"	"	"	"	"
t_{PLH2}		176	"	"	4.5V	IN	"	"		"	"							"	G2A to Y6	"	31	"	41	"	31	"
t_{PLH2}		177	"	"	"	GND	IN	"		"	"							"	G2B to Y6	"	"	"	"	"	"	"
t_{PLH2}		178	"	"	"	"	GND	IN		"	"							"	G1 to Y6	"	"	"	"	"	"	"
t_{PLH1}		179	IN	"	"	"	"	4.5V	OUT	"								"	A to Y7	"	35	"	47	"	35	"
t_{PLH1}		180	4.5V	IN	"	"	"	"	"	"								"	B to Y7	"	"	"	"	"	"	"
t_{PLH1}		181	"	4.5V	IN	"	"	"	"	"								"	C to Y7	"	"	"	"	"	"	"
t_{PLH2}		182	"	"	4.5V	IN	"	"	"	"								"	G2A to Y7	"	31	"	41	"	31	"
t_{PLH2}		183	"	"	"	GND	IN	"	"	"								"	G2B to Y7	"	"	"	"	"	"	"
t_{PLH2}		184	"	"	"	GND	GND	IN	"	"								"	G1 to Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 4)	185	GND	GND	GND	GND	GND	IN		GND						OUT	OUT	4.5V	Y0	3	15	3	20	3	15	ns
		186	4.5V	GND	"	"	"	"		"								"	Y1	"	"	"	"	"	"	"
		187	GND	4.5V	"	"	"	"		"					OUT			"	Y2	"	"	"	"	"	"	"
		188	4.5V	4.5V	"	"	"	"		"				OUT				"	Y3	"	"	"	"	"	"	"
		189	GND	GND	4.5V	"	"	"		"			OUT					"	Y4	"	"	"	"	"	"	"
		190	4.5V	GND	"	"	"	"		"		OUT						"	Y5	"	"	"	"	"	"	"
		191	GND	4.5V	"	"	"	"		"	OUT							"	Y6	"	"	"	"	"	"	"
		192	4.5V	4.5V	"	"	"	"	OUT	"								"	Y7	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 4)	193	GND	GND	GND	"	"	"		"						OUT	OUT	"	Y0	"	"	"	"	"	"	"
		194	4.5V	GND	"	"	"	"		"					OUT			"	Y1	"	"	"	"	"	"	"
		195	GND	4.5V	"	"	"	"		"				OUT				"	Y2	"	"	"	"	"	"	"
		196	4.5V	4.5V	"	"	"	"		"			OUT					"	Y3	"	"	"	"	"	"	"
		197	GND	GND	4.5V	"	"	"		"								"	Y4	"	"	"	"	"	"	"
		198	4.5V	GND	"	"	"	"		"		OUT						"	Y5	"	"	"	"	"	"	"
		199	GND	4.5V	"	"	"	"		"	OUT							"	Y6	"	"	"	"	"	"	"
		200	4.5V	4.5V	"	"	"	"	OUT	"								"	Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA							1/								GND	G1		1.5					V
		2		1mA						"								"	A1		"					"
		3			1mA					"								"	B1		"					"
		4								"							1mA	"	G2		"					"
		5								"						1mA		"	A2		"					"
		6								"					1mA			"	B2		"					"
V_{IC}(neg)		7	-1mA							GND								1/	G1		-1.5					"
		8		-1mA						"								"	A1		"					"
		9			-1mA					"								"	B1		"					"
		10								"							-1mA	"	G2		"					"
		11								"						-1mA		"	A2		"					"
		12								"					-1mA			"	B2		"					"
I_{CC}	3005	13	6.0V	6.0V	6.0V					"					6.0V	6.0V	6.0V	6.0V	V_{CC}		0.2		20			µA
V_{OH3}	3006	14	4.2V	GND	GND	-20µA				"					GND	GND		"	1Y0	5.95		5.95		5.95		V
		15	"	"	"		-20µA			"					"	"		"	1Y1	"		"		"		"
		16	"	"	"			-20µA		"					"	"		"	1Y2	"		"		"		"
		17	"	"	"				-20µA	"					"	"		"	1Y3	"		"		"		"
		18		"	"					"				-20µA	"	"	4.2V	"	2Y0	"		"		"		"
		19		"	"					"			-20µA		"	"	"	"	2Y1	"		"		"		"
		20		"	"					"		-20µA			"	"	"	"	2Y2	"		"		"		"
		21		"	"					"	-20µA				"	"	"	"	2Y3	"		"		"		"
V_{OH5}	3006	22	4.2V	"	"	-5.2mA				"					"	"		"	1Y0	5.48		5.2		5.48		"
		23	"	"	"		-5.2mA			"					"	"		"	1Y1	"		"		"		"
		24	"	"	"			-5.2mA		"					"	"		"	1Y2	"		"		"		"
		25	"	"	"				-5.2mA	"					"	"		"	1Y3	"		"		"		"
		26		"	"					"				-5.2mA	"	"	4.2V	"	2Y0	"		"		"		"
		27		"	"					"			-5.2mA		"	"	"	"	2Y1	"		"		"		"
		28		"	"					"		-5.2mA			"	"	"	"	2Y2	"		"		"		"
		29		"	"					"	-5.2mA				"	"	"	"	2Y3	"		"		"		"
V_{OL3}	3007	30	GND	"	"	20µA				"					"	"	GND	"	1Y0		0.05		0.05		0.05	"
		31	"	4.2V	"		20µA			"					"	"	"	"	1Y1		"		"		"	"
		32	"	GND	4.2V			20µA		"					"	"	"	"	1Y2		"		"		"	"
		33	"	4.2V	4.2V				20µA	"					"	"	"	"	1Y3		"		"		"	"
		34	"	GND	GND					"				20µA	"	"	"	"	2Y0		"		"		"	"
		35	"	"	"					"			20µA		"	4.2V	"	"	2Y1		"		"		"	"
		36	"	"	"					"		20µA			4.2V	GND	"	"	2Y2		"		"		"	"
		37	"	"	"					"	20µA				4.2V	4.2V	"	"	2Y3		"		"		"	"
V_{OL5}	3007	38	"	"	"	5.2mA				"					GND	GND	"	"	1Y0		0.26		0.4		0.26	"
		39	"	4.2V	"		5.2mA			"					"	"	"	"	1Y1		"		"		"	"
		40	"	GND	4.2V			5.2mA		"					"	"	"	"	1Y2		"		"		"	"
		41	"	4.2V	4.2V				5.2mA	"					"	"	"	"	1Y3		"		"		"	"
		42	"	GND	GND					"				5.2mA	"	"	"	"	2Y0		"		"		"	"
		43	"	"	"					"			5.2mA		"	4.2V	"	"	2Y1		"		"		"	"
		44	"	"	"					"		5.2mA			4.2V	GND	"	"	2Y2		"		"		"	"
		45	"	"	"					"	5.2mA				4.2V	4.2V	"	"	2Y3		"		"		"	"

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	46	4.0V	GND	GND	GND	GND	GND	GND	GND					GND	GND		4.0V	1Y0	-10	-120	-10	-120	-10	-120	mA
		47	"	"	"					"					"	"		"	1Y1	"	"	"	"	"	"	"
		48	"	"	"					"					"	"		"	1Y2	"	"	"	"	"	"	"
		49	"	"	"				GND	"					"	"		"	1Y3	"	"	"	"	"	"	"
		50		"	"					"				GND	"	"	4.0V	"	2Y0	"	"	"	"	"	"	"
		51		"	"					"			GND		"	"	"	"	2Y1	"	"	"	"	"	"	"
		52		"	"					"		GND			"	"	"	"	2Y2	"	"	"	"	"	"	"
		53		"	"					"	GND				"	"	"	"	2Y3	"	"	"	"	"	"	"
I_{IH}	3010	54	6.0V	"	"					"					"	"	GND	6.0V	G1		.05		0.1			µA
		55	GND	6.0V	"					"					"	"	"	"	A1		"		"			"
		56	"	GND	6.0V					"					"	"	"	"	B1		"		"			"
		57	"	"	GND					"					"	"	6.0V	"	G2		"		"			"
		58	"	"	"					"					"	6.0V	GND	"	A2		"		"			"
		59	"	"	"					"					6.0V	GND	GND	"	B2		"		"			"
I_{IL}	3009	60	GND	6.0V	6.0V					"					"	6.0V	6.0V	6.0V	G1		-.05		-0.1			µA
		61	6.0V	GND	6.0V					"					"	"	"	"	A1		"		"			"
		62	"	6.0V	GND					"					"	"	"	"	B1		"		"			"
		63	"	"	6.0V					"					"	"	GND	"	G2		"		"			"
		64	"	"	"					"					"	GND	6.0V	"	A2		"		"			"
		65	"	"	"					"					GND	6.0V	6.0V	"	B2		"		"			"
																				Subgroup 4 T_C = +25°C						
																				Min	Max					
C_i	3012	66	2/							GND								GND	G1	10						pF
		67		2/						"								"	A1	"						"
		68			2/					"								"	B1	"						"
		69								"							2/	"	G2	"						"
		70								"						2/		"	A2	"						"
		71								"					2/			"	B2	"						"
																				Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table tests 3/	3014	72	A	A	A	H	H	H	H	GND	H	H	H	H	A	A	A	4.5V	4/							
		73	"	B	A	"	"	"	"	"	"	"	"	"	A	B	"	"									
		74	"	A	B	"	"	"	"	"	"	"	"	"	B	A	"	"									
		75	"	B	B	"	"	"	"	"	"	"	"	"	"	B	"	"									
		76	B	B	A	L	"	"	"	"	"	"	"	L	"	B	B	"									
		77	"	A	B	H	L	"	"	"	"	"	L	H	"	A	"	"									
		78	"	B	A	"	H	L	"	"	"	L	H	"	A	B	"	"									
		79	"	A	A	"	H	H	L	"	L	H	H	"	A	A	"	"									

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL2}	3003 (Fig. 4)	80	IN	GND	GND	OUT				GND					GND	GND	GND	4.5V	G1 to 1Y0	6	31	6	41	6	31	ns
t_{PHL1}		81	GND	IN	GND	"				"					"	"	"	"	A1 to 1Y0	"	39	"	51	"	39	"
t_{PHL1}		82	GND	GND	IN	"				"					"	"	"	"	B1 to 1Y0	"	39	"	51	"	39	"
t_{PHL2}		83	IN	4.5V	GND		OUT			"					"	"	"	"	G1 to 1Y1	"	31	"	41	"	31	"
t_{PHL1}		84	GND	IN	GND		"			"					"	"	"	"	A1 to 1Y1	"	39	"	51	"	39	"
t_{PHL1}		85	GND	4.5V	IN		"			"					"	"	"	"	B1 to 1Y1	"	39	"	51	"	39	"
t_{PHL2}		86	IN	GND	4.5V			OUT		"					"	"	"	"	G1 to 1Y2	"	31	"	41	"	31	"
t_{PHL1}		87	GND	IN	4.5V			"		"					"	"	"	"	A1 to 1Y2	"	39	"	51	"	39	"
t_{PHL1}		88	GND	GND	IN			"		"					"	"	"	"	B1 to 1Y2	"	39	"	51	"	39	"
t_{PHL2}		89	IN	4.5V	4.5V				OUT	"					"	"	"	"	G1 to 1Y3	"	31	"	41	"	31	"
t_{PHL1}		90	GND	IN	4.5V				"	"					"	"	"	"	A1 to 1Y3	8	39	8	51	8	39	"
t_{PHL1}		91	"	4.5V	IN				"	"					"	"	"	"	B1 to 1Y3	8	39	8	51	8	39	"
t_{PHL2}		92	"	GND	GND					"				OUT	"	"	IN	"	G2 to 2Y0	6	31	6	41	6	31	"
t_{PHL1}		93	"	"	"					"				"	"	IN	GND	"	A2 to 2Y0	"	39	"	51	"	39	"
t_{PHL1}		94	"	"	"					"				"	IN	GND	GND	"	B2 to 2Y0	"	39	"	51	"	39	"
t_{PHL2}		95	"	"	"					"			OUT		GND	4.5V	IN	"	G2 to 2Y1	"	31	"	41	"	31	"
t_{PHL1}		96	"	"	"					"			"		GND	IN	GND	"	A2 to 2Y1	"	39	"	51	"	39	"
t_{PHL1}		97	"	"	"					"			"		IN	4.5V	GND	"	B2 to 2Y1	"	39	"	51	"	39	"
t_{PHL2}		98	"	"	"					"		OUT			4.5V	GND	IN	"	G2 to 2Y2	"	31	"	41	"	31	"
t_{PHL1}		99	"	"	"					"		"			4.5V	IN	GND	"	A2 to 2Y2	"	39	"	51	"	39	"
t_{PHL1}		100	"	"	"					"		"			IN	GND	GND	"	B2 to 2Y2	"	39	"	51	"	39	"
t_{PHL2}		101	"	"	"					"	OUT				4.5V	4.5V	IN	"	G2 to 2Y3	"	31	"	41	"	31	"
t_{PHL1}		102	"	"	"					"	"				4.5V	IN	GND	"	A2 to 2Y3	8	39	8	51	8	39	"
t_{PHL1}		103	"	"	"					"	"				IN	4.5V	"	"	B2 to 2Y3	8	39	8	51	8	39	"
t_{PLH2}		104	IN	"	"	OUT				"					GND	GND	"	"	G1 to 1Y0	6	31	6	41	6	31	"
t_{PLH1}		105	GND	IN	"	"				"					"	"	"	"	A1 to 1Y0	"	39	"	51	"	39	"
t_{PLH1}		106	GND	GND	IN	"				"					"	"	"	"	B1 to 1Y0	"	39	"	51	"	39	"
t_{PLH2}		107	IN	4.5V	GND		OUT			"					"	"	"	"	G1 to 1Y1	"	31	"	41	"	31	"
t_{PLH1}		108	GND	IN	GND		"			"					"	"	"	"	A1 to 1Y1	"	39	"	51	"	39	"
t_{PLH1}		109	GND	4.5V	IN					"					"	"	"	"	B1 to 1Y1	"	39	"	51	"	39	"
t_{PLH2}		110	IN	GND	4.5V			OUT		"					"	"	"	"	G1 to 1Y2	"	31	"	41	"	31	"
t_{PLH1}		111	GND	IN	4.5V			"		"					"	"	"	"	A1 to 1Y2	"	39	"	51	"	39	"
t_{PLH1}		112	GND	GND	IN			"		"					"	"	"	"	B1 to 1Y2	"	39	"	51	"	39	"
t_{PLH2}		113	IN	4.5V	4.5V				OUT	"					"	"	"	"	G1 to 1Y3	"	31	"	41	"	31	"
t_{PLH1}		114	GND	IN	4.5V				"	"					"	"	"	"	A1 to 1Y3	8	39	8	51	8	39	"
t_{PLH1}		115	"	GND	IN				"	"					"	"	"	"	B1 to 1Y3	8	39	8	51	8	39	"
t_{PLH2}		116	"	"	GND					"				OUT	"	"	IN	"	G2 to 2Y0	6	31	6	41	6	31	"
t_{PLH1}		117	"	"	"					"				"	"	IN	GND	"	A2 to 2Y0	"	39	"	51	"	39	"
t_{PLH1}		118	"	"	"					"				"	IN	GND	GND	"	B2 to 2Y0	"	39	"	51	"	39	"
t_{PLH2}		119	"	"	"					"			OUT		GND	4.5V	IN	"	G2 to 2Y1	"	31	"	41	"	31	"
t_{PLH1}		120	"	"	"					"			"		GND	IN	GND	"	A2 to 2Y1	"	39	"	51	"	39	"
t_{PLH1}		121	"	"	"					"			"		IN	4.5V	GND	"	B2 to 2Y1	"	39	"	51	"	39	"
t_{PLH2}		122	"	"	"					"		OUT			4.5V	GND	IN	"	G2 to 2Y2	"	31	"	41	"	31	"
t_{PLH1}		123	"	"	"					"		"			4.5V	IN	GND	"	A2 to 2Y2	"	39	"	51	"	39	"
t_{PLH1}		124	"	"	"					"		"			IN	GND	GND	"	B2 to 2Y2	"	39	"	51	"	39	"
t_{PLH2}		125	"	"	"					"	OUT				4.5V	4.5V	IN	"	G2 to 2Y3	"	31	"	41	"	31	"
t_{PLH1}		126	"	"	"					"	"				4.5V	IN	GND	"	A2 to 2Y3	8	39	8	51	8	39	"
t_{PLH1}		127	"	"	"					"	"				IN	4.5V	"	"	B2 to 2Y3	8	39	8	51	8	39	"

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Case 2 / E and F / Test no.	Terminal conditions 1/ 2 / 1 / G1	3 / 2 / A1	4 / 3 / B1	5 / 4 / 1Y0	7 / 5 / 1Y1	8 / 6 / 1Y2	9 / 7 / 1Y3	10 / 8 / GND	12 / 9 / 2Y3	13 / 10 / 2Y2	14 / 11 / 2Y1	15 / 12 / 2Y0	17 / 13 / B2	18 / 14 / A2	19 / 15 / G2	20 / 16 / V_{CC}	Measured terminal	Test limits Subgroup 9 T_C = +25°C 5/ Min	Subgroup 9 Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C 5/ Min	Subgroup 11 Max	Unit
t_{THL}	3003 (Fig. 4)	128	IN	GND	GND					GND					GND	GND	GND	4.5V	1Y0	3	15	3	20	3	15	ns
		129	"	4.5V	"					"					"	"	"	"	1Y1	"	"	"	"	"	"	"
		130	"	GND	4.5V					"					"	"	"	"	1Y2	"	"	"	"	"	"	"
		131	"	4.5V	4.5V					"					"	"	"	"	1Y3	"	"	"	"	"	"	"
		132	GND	GND	GND					"					"	GND	IN	"	2Y0	"	"	"	"	"	"	"
		133	"	"	"					"					"	4.5V	"	"	2Y1	"	"	"	"	"	"	"
		134	"	"	"					"					4.5V	GND	"	"	2Y2	"	"	"	"	"	"	"
		135	"	"	"					"					4.5V	4.5V	"	"	2Y3	"	"	"	"	"	"	"
t_{TLH}	3003 (Fig. 4)	136	IN	"	"					"					GND	GND	GND	"	1Y0	"	"	"	"	"	"	"
		137	"	4.5V	"					"					"	"	"	"	1Y1	"	"	"	"	"	"	"
		138	"	GND	4.5V					"					"	"	"	"	1Y2	"	"	"	"	"	"	"
		139	"	4.5V	4.5V					"					"	"	"	"	1Y3	"	"	"	"	"	"	"
		140	GND	GND	GND					"					"	GND	IN	"	2Y0	"	"	"	"	"	"	"
		141	"	"	"					"					"	4.5V	"	"	2Y1	"	"	"	"	"	"	"
		142	"	"	"					"					4.5V	GND	"	"	2Y2	"	"	"	"	"	"	"
		143	"	"	"					"					4.5V	4.5V	"	"	2Y3	"	"	"	"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Terminal conditions 1/																								Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		K and L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1												1/						1mA				1mA	1mA	GND	A		1.5					V
		2												"												"	B		"					"
		3												"									1mA			"	C		"					"
		4												"								1mA				"	D		"					"
		5												"							1mA					"	G2		"					"
		6												"						1mA						"	G1		"					"
V_{IC}(neg)		7												GND											-1mA	1/	A		-1.5					"
		8												"												"	B		"					"
		9												"									-1mA	-1mA		"	C		"					"
		10												"								-1mA				"	D		"					"
		11												"							-1mA					"	G2		"					"
		12												"						-1mA						"	G1		"					"
I_{CC}	3005	13												"						6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	V_{CC}		0.2		20			µA
V_{OH3}	3006	14	-20µA											"						4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	"	Y0	5.95		5.95		5.95		V
		15		-20µA										"						"	"	"	"	"	"	"	Y1	"		"		"		"
		16			-20µA									"						"	"	"	"	"	"	"	Y2	"		"		"		"
		17				-20µA								"						"	"	"	"	"	"	"	Y3	"		"		"		"
		18					-20µA							"						"	"	"	"	"	"	"	Y4	"		"		"		"
		19						-20µA						"						"	"	"	"	"	"	"	Y5	"		"		"		"
		20							-20µA					"						"	"	"	"	"	"	"	Y6	"		"		"		"
		21								-20µA				"						"	"	"	"	"	"	"	Y7	"		"		"		"
		22									-20µA			"						"	"	"	"	"	"	"	Y8	"		"		"		"
		23										-20µA		"						"	"	"	"	"	"	"	Y9	"		"		"		"
		24											-20µA	"						"	"	"	"	"	"	"	Y10	"		"		"		"
		25												"	-20µA					"	"	"	"	"	"	"	Y11	"		"		"		"
		26												"		-20µA				"	"	"	"	"	"	"	Y12	"		"		"		"
		27												"			-20µA			"	"	"	"	"	"	"	Y13	"		"		"		"
		28												"				-20µA		"	"	"	"	"	"	"	Y14	"		"		"		"
		29												"					-20µA	"	"	"	"	"	"	"	Y15	"		"		"		"
V_{OH5}	3006	30	-5.2mA											"						"	"	"	"	"	"	"	Y0	5.48		5.2		5.48		"
		31		-5.2mA										"						"	"	"	"	"	"	"	Y1	"		"		"		"
		32			-5.2mA									"						"	"	"	"	"	"	"	Y2	"		"		"		"
		33				-5.2mA								"						"	"	"	"	"	"	"	Y3	"		"		"		"
		34					-5.2mA							"						"	"	"	"	"	"	"	Y4	"		"		"		"
		35						-5.2mA						"						"	"	"	"	"	"	"	Y5	"		"		"		"
		36							-5.2mA					"						"	"	"	"	"	"	"	Y6	"		"		"		"
		37								-5.2mA				"						"	"	"	"	"	"	"	Y7	"		"		"		"
		38									-5.2mA			"						"	"	"	"	"	"	"	Y8	"		"		"		"
		39										-5.2mA		"						"	"	"	"	"	"	"	Y9	"		"		"		"
		40											-5.2mA	"						"	"	"	"	"	"	"	Y10	"		"		"		"
		41												"	-5.2mA					"	"	"	"	"	"	"	Y11	"		"		"		"
		42												"		-5.2mA				"	"	"	"	"	"	"	Y12	"		"		"		"
		43												"			-5.2mA			"	"	"	"	"	"	"	Y13	"		"		"		"
		44												"				-5.2mA		"	"	"	"	"	"	"	Y14	"		"		"		"
		45												"					-5.2mA	"	"	"	"	"	"	"	Y15	"		"		"		"

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																									Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		K and L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{OL3}	3007	46	20µA											GND						GND	GND	GND	GND	GND	GND	6.0V	Y0		0.05		0.05		0.05	V
		47		20µA										"						"	"	"	"	GND	4.2V	"	Y1		"		"		"	"
		48			20µA									"						"	"	"	"	4.2V	GND	"	Y2		"		"		"	"
		49				20µA								"						"	"	"	"	4.2V	4.2V	"	Y3		"		"		"	"
		50					20µA							"						"	"	"	4.2V	GND	GND	"	Y4		"		"		"	"
		51						20µA						"						"	"	"	"	GND	4.2V	"	Y5		"		"		"	"
		52							20µA					"						"	"	"	"	4.2V	GND	"	Y6		"		"		"	"
		53								20µA				"						"	"	"	"	4.2V	4.2V	"	Y7		"		"		"	"
		54									20µA			"						"	"	4.2V	GND	GND	GND	"	Y8		"		"		"	"
		55										20µA		"						"	"	"	"	GND	4.2V	"	Y9		"		"		"	"
		56											20µA	"						"	"	"	"	4.2V	GND	"	Y10		"		"		"	"
		57												"	20µA					"	"	"	"	4.2V	4.2V	"	Y11		"		"		"	"
		58												"		20µA				"	"	"	4.2V	GND	GND	"	Y12		"		"		"	"
		59												"			20µA			"	"	"	"	GND	4.2V	"	Y13		"		"		"	"
		60												"				20µA		"	"	"	"	4.2V	GND	"	Y14		"		"		"	"
		61												"					20µA	"	"	"	"	4.2V	4.2V	"	Y15		"		"		"	"
V_{OL5}	3007	62	5.2mA											"						"	"	GND	GND	GND	GND	"	Y0		0.26		0.4		0.26	"
		63		5.2mA										"						"	"	"	"	GND	4.2V	"	Y1		"		"		"	"
		64			5.2mA									"						"	"	"	"	4.2V	GND	"	Y2		"		"		"	"
		65				5.2mA								"						"	"	"	"	4.2V	4.2V	"	Y3		"		"		"	"
		66					5.2mA							"						"	"	"	4.2V	GND	GND	"	Y4		"		"		"	"
		67						5.2mA						"						"	"	"	"	GND	4.2V	"	Y5		"		"		"	"
		68							5.2mA					"						"	"	"	"	4.2V	GND	"	Y6		"		"		"	"
		69								5.2mA				"						"	"	"	"	4.2V	4.2V	"	Y7		"		"		"	"
		70									5.2mA			"						"	"	4.2V	GND	GND	GND	"	Y8		"		"		"	"
		71										5.2mA		"						"	"	"	"	GND	4.2V	"	Y9		"		"		"	"
		72											5.2mA	"						"	"	"	"	4.2V	GND	"	Y10		"		"		"	"
		73												"	5.2mA					"	"	"	"	4.2V	4.2V	"	Y11		"		"		"	"
		74												"		5.2mA				"	"	"	4.2V	GND	GND	"	Y12		"		"		"	"
		75												"			5.2mA			"	"	"	"	GND	4.2V	"	Y13		"		"		"	"
		76												"				5.2mA		"	"	"	"	4.2V	GND	"	Y14		"		"		"	"
		77												"					5.2mA	"	"	"	"	4.2V	4.2V	"	Y15		"		"		"	"
I_{OS4}	3011	78	GND											"						4.0V	4.0V	GND	GND	GND	GND	4.0V	Y0	-10	-120	-10	-120	-10	-120	mA
		79		GND										"						"	"	"	"	"	"	"	Y1	"	"	"	"	"	"	"
		80			GND									"						"	"	"	"	"	"	"	Y2	"	"	"	"	"	"	"
		81				GND								"						"	"	"	"	"	"	"	Y3	"	"	"	"	"	"	"
		82					GND							"						"	"	"	"	"	"	"	Y4	"	"	"	"	"	"	"
		83						GND						"						"	"	"	"	"	"	"	Y5	"	"	"	"	"	"	"
		84							GND					"						"	"	"	"	"	"	"	Y6	"	"	"	"	"	"	"
		85								GND				"						"	"	"	"	"	"	"	Y7	"	"	"	"	"	"	"
		86									GND			"						"	"	"	"	"	"	"	Y8	"	"	"	"	"	"	"
		87										GND		"						"	"	"	"	"	"	"	Y9	"	"	"	"	"	"	"
		88											GND	"						"	"	"	"	"	"	"	Y10	"	"	"	"	"	"	"
		89												"	GND					"	"	"	"	"	"	"	Y11	"	"	"	"	"	"	"
		90												"		GND				"	"	"	"	"	"	"	Y12	"	"	"	"	"	"	"
		91												"			GND			"	"	"	"	"	"	"	Y13	"	"	"	"	"	"	"
		92												"				GND		"	"	"	"	"	"	"	Y14	"	"	"	"	"	"	"
		93												"					GND	"	"	"	"	"	"	"	Y15	"	"	"	"	"	"	"
I_{IH}	3010	94												"						GND	GND	GND	GND	GND	6.0V	6.0V	A		0.05		0.1			µA
		95												"						"	"	"	GND	6.0V	GND	"	B		"		"			"
		96												"						"	"	"	6.0V	GND	"	"	C		"		"			"
		97												"						"	"	6.0V	GND	"	"	"	D		"		"			"
		98												"						6.0V	"	GND	"	"	"	"	G1		"		"			"
		99												"						GND	6.0V	"	"	"	"	"	G2		"		"			"
I_{IL}	3009	100												"						"	GND	GND	"	"	"	"	A		-0.05		-0.1			"
		101												"						"	"	"	"	"	"	"	B		"		"			"
		102												"						"	"	"	"	"	"	"	C		"		"			"
		103												"						"	"	"	"	"	"	"	D		"		"			"
		104												"						6.0V	"	"	"	"	"	"	G2		"		"			"
		105												"						GND	6.0V	"	"	"	"	"	G1		"		"			"

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																								Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 4 T_C = +25°C					
		K and L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24							
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}		Min	Max					
C_i	3012	106												GND												GND	A		10					pF
		107												"												"	B		"					"
		108												"												"	C		"					"
		109												"												"	D		"					"
		110												"						2/	2/	2/	2/	2/	2/		"	G1		"					"
		111												"												"	G2		"					"
																												Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																												Min	Max	Min	Max	Min	Max	
Truth table tests 3/	3014	112	L	H	H	H	H	H	H	H	H	H	H	H	GND	H	H	H	H	H	B	B	B	B	B	B	4.5V	4/						
		113	H	L	L	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		114	"	H	H	H	L	L	L	L	L	L	L	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		115	"	"	"	"	H	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		116	"	"	"	"	"	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		117	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"						
		118	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"						
		119	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"						
		120	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"						
		121	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		122	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		123	"	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		124	"	"	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"	"						
		125	"	"	"	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"						
		126	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"						
		127	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"	"						
		128	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"	"	"	"	"	"						
		129	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	A	A	X	X	X	X	X	X	"	"						
		130	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	A	B	"	"	"	"	"	"	"	"						
																												Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C= +125°C		Subgroup 11 T_C = -55°C 5/		
																												Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 4)	131	OUT											GND						GND	GND	GND	GND	GND	IN	4.5V	A to Y0	6	33	6	44	6	33	ns
		132		OUT										"						"	"	"	GND	IN	4.5V	"	B to Y1	"	"	"	"	"	"	"
		133			OUT									"						"	"	"	IN	4.5V	GND	"	C to Y2	"	"	"	"	"	"	"
		134				OUT								"						"	"	IN	GND	4.5V	4.5V	"	D to Y3	"	"	"	"	"	"	"
		135					OUT							"						"	"	GND	4.5V	GND	IN	"	A to Y4	"	"	"	"	"	"	"
		136						OUT						"						"	"	"	"	IN	4.5V	"	B to Y5	"	"	"	"	"	"	"
		137							OUT					"						"	"	"	"	4.5V	IN	"	A to Y6	"	"	"	"	"	"	"
		138								OUT				"						"	"	IN	"	4.5V	4.5V	"	D to Y7	"	"	"	"	"	"	"
		139									OUT			"						"	"	4.5V	GND	GND	IN	"	A to Y8	"	"	"	"	"	"	"
		140										OUT		"						"	"	"	GND	IN	4.5V	"	B to Y9	"	"	"	"	"	"	"
		141											OUT	"						"	"	"	IN	4.5V	GND	"	C to Y10	"	"	"	"	"	"	"
		142												"	OUT					"	"	"	IN	4.5V	4.5V	"	C to Y11	"	"	"	"	"	"	"
		143												"		OUT				"	"	"	4.5V	GND	IN	"	A to Y12	"	"	"	"	"	"	"
		144												"			OUT			"	"	"	"	IN	4.5V	"	B to Y13	"	"	"	"	"	"	"
		145												"				OUT		"	"	"	"	4.5V	IN	"	A to Y14	"	"	"	"	"	"	"
t_{PHL2}	3003 (Fig. 4)	146	OUT											"					OUT	IN	"	GND	GND	GND	GND	"	G1 to Y0	"	32	"	42	"	32	"
		147												"						GND	IN	4.5V	4.5V	4.5V	4.5V	"	G2 to Y15	"	32	"	42	"	32	"
t_{PLH1}	3003 (Fig. 4)	148		OUT										"						"	GND	GND	GND	GND	IN	"	A to Y1	"	33	"	44	"	33	"
		149			OUT									"						"	"	"	"	GND	GND	"	B to Y2	"	"	"	"	"	"	"
		150				OUT								"						"	"	"	IN	GND	4.5V	"	B to Y3	"	"	"	"	"	"	"
		151					OUT							"						"	"	"	"	GND	GND	"	C to Y4	"	"	"	"	"	"	"
		152						OUT						"						"	"	"	"	4.5V	4.5V	"	C to Y5	"	"	"	"	"	"	"
		153							OUT					"						"	"	"	"	4.5V	GND	"	C to Y6	"	"	"	"	"	"	"
		154								OUT				"						"	"	4.5V	4.5V	IN	IN	"	A to Y7	"	"	"	"	"	"	"
		155									OUT			"						"	"	IN	GND	GND	GND	"	D to Y8	"	"	"	"	"	"	"
		156										OUT		"						"	"	"	"	GND	4.5V	"	D to Y9	"	"	"	"	"	"	"
		157											OUT	"						"	"	"	"	4.5V	GND	"	D to Y10	"	"	"	"	"	"	"
		158												"	OUT					"	"	"	"	4.5V	4.5V	"	D to Y11	"	"	"	"	"	"	"
		159												"		OUT				"	"	"	4.5V	GND	GND	"	D to Y12	"	"	"	"	"	"	"
		160												"			OUT			"	"	4.5V	4.5V	GND	IN	"	A to Y13	"	"	"	"	"	"	"
		161												"				OUT		"	"	4.5V	IN	4.5V	GND	"	C to Y14	"	"	"	"	"	"	"
		162												"					OUT	"	"	IN	4.5V	4.5V	4.5V	"	D to Y15	"	"	"	"	"	"	"
t_{PLH2}	3003 (Fig. 4)	163	OUT											"						IN	"	GND	GND	GND	GND	"	G1 to Y0	"	32	"	42	"	32	"
		164												"					OUT	GND	IN	4.5V	4.5V	4.5V	4.5V	"	G2 to Y15	"	32	"	42	"	32	"

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Terminal conditions 1/																								Measured terminal	Test limits						Unit
		Case 3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		K and L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24								
		Test no.	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	GND	Y11	Y12	Y13	Y14	Y15	G1	G2	D	C	B	A	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 4)	165	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	OUT	IN	GND	GND	GND	GND	GND	4.5V	Y0	3	15	3	20	3	15	ns
		166												"						"	"	"	"	GND	4.5V	"	Y1	"	"	"	"	"	"	"
		167												"						"	"	"	"	4.5V	GND	"	Y2	"	"	"	"	"	"	"
		168												"						"	"	"	"	4.5V	4.5V	"	Y3	"	"	"	"	"	"	"
		169												"						"	"	"	4.5V	GND	GND	"	Y4	"	"	"	"	"	"	"
		170												"						"	"	"	"	GND	4.5V	"	Y5	"	"	"	"	"	"	"
		171												"						"	"	"	"	4.5V	GND	"	Y6	"	"	"	"	"	"	"
		172												"						"	"	"	"	4.5V	4.5V	"	Y7	"	"	"	"	"	"	"
		173												"						"	"	4.5V	GND	GND	GND	"	Y8	"	"	"	"	"	"	"
		174												"						"	"	"	"	GND	4.5V	"	Y9	"	"	"	"	"	"	"
		175												"						"	"	"	"	4.5V	GND	"	Y10	"	"	"	"	"	"	"
		176												"						"	"	"	"	4.5V	4.5V	"	Y11	"	"	"	"	"	"	"
		177												"						"	"	"	4.5V	GND	GND	"	Y12	"	"	"	"	"	"	"
		178												"						"	"	"	"	GND	4.5V	"	Y13	"	"	"	"	"	"	"
		179												"						"	"	"	"	4.5V	GND	"	Y14	"	"	"	"	"	"	"
		180												"						"	"	"	"	4.5V	4.5V	"	Y15	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 4)	181	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	OUT	"	"	GND	GND	GND	GND	"	Y0	"	"	"	"	"	"	"
		182												"						"	"	"	"	GND	4.5V	"	Y1	"	"	"	"	"	"	"
		183												"						"	"	"	"	4.5V	GND	"	Y2	"	"	"	"	"	"	"
		184												"						"	"	"	"	4.5V	4.5V	"	Y3	"	"	"	"	"	"	"
		185												"						"	"	"	4.5V	GND	GND	"	Y4	"	"	"	"	"	"	"
		186												"						"	"	"	"	GND	4.5V	"	Y5	"	"	"	"	"	"	"
		187												"						"	"	"	"	4.5V	GND	"	Y6	"	"	"	"	"	"	"
		188												"						"	"	"	"	4.5V	4.5V	"	Y7	"	"	"	"	"	"	"
		189												"						"	"	4.5V	GND	GND	GND	"	Y8	"	"	"	"	"	"	"
		190												"						"	"	"	"	GND	4.5V	"	Y9	"	"	"	"	"	"	"
		191												"						"	"	"	"	4.5V	GND	"	Y10	"	"	"	"	"	"	"
		192												"						"	"	"	"	4.5V	4.5V	"	Y11	"	"	"	"	"	"	"
		193												"						"	"	"	4.5V	GND	GND	"	Y12	"	"	"	"	"	"	"
		194												"						"	"	"	"	GND	4.5V	"	Y13	"	"	"	"	"	"	"
		195												"						"	"	"	"	4.5V	GND	"	Y14	"	"	"	"	"	"	"
		196												"						"	"	"	"	4.5V	4.5V	"	Y15	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 52.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA							1/								GND	A		1.5					V
		2		1mA						"								"	B		"					"
		3			1mA					"								"	C		"					"
		4				1mA				"								"	$\overline{G2A}$		"					"
		5					1mA			"								"	G2B		"					"
		6						1mA		"								"	G1		"					"
V_{IC}(neg)		7	-1mA							GND								1/	A		-1.5					"
		8		-1mA						"								"	B		"					"
		9			-1mA					"								"	C		"					"
		10				-1mA				"								"	$\overline{G2A}$		"					"
		11					-1mA			"								"	G2B		"					"
		12						-1mA		"								"	G1		"					"
I_{CC}	3005	13	5.5V	5.5V	5.5V	5.5V	5.5V	5.5V		"								5.5V	V_{CC}		0.2		20			µA
$I_{CC}\Delta$		14	0.8V	0.8V	0.8V	0.8V	0.8V	2.4V		"								"	"		3.0		3.0		3.0	mA
		15	2.4V	0.8V	0.8V	0.8V	"	5.5V		"								"	"		"		"		"	"
		16				2.4V	"	0.8V		"								"	"		"		"		"	"
V_{OH6}	3006	17	0.8V	0.8V	0.8V	2.0V	0.8V	0.8V		"							-20µA	4.5V	Y0	4.40		4.40		4.40		V
		18	"	"	"	"	"	"		"						-20µA		"	Y1	"		"		"		"
		19	"	"	"	"	"	"		"					-20µA			"	Y2	"		"		"		"
		20	"	"	"	"	"	"		"				-20µA				"	Y3	"		"		"		"
		21	"	"	"	"	"	"		"			-20µA					"	Y4	"		"		"		"
		22	"	"	"	"	"	"		"		-20µA						"	Y5	"		"		"		"
		23	"	"	"	"	"	"		"	-20µA							"	Y6	"		"		"		"
		24	"	"	"	"	"	"	-20µA	"								"	Y7	"		"		"		"
V_{OH7}	3006	25	"	"	"	"	"	"		"							-4.0mA	5.5V	Y0	3.98		3.70		3.98		"
		26	"	"	"	"	"	"		"						-4.0mA		"	Y1	"		"		"		"
		27	"	"	"	"	"	"		"					-4.0mA			"	Y2	"		"		"		"
		28	"	"	"	"	"	"		"				-4.0mA				"	Y3	"		"		"		"
		29	"	"	"	"	"	"		"			-4.0mA					"	Y4	"		"		"		"
		30	"	"	"	"	"	"		"		-4.0mA						"	Y5	"		"		"		"
		31	"	"	"	"	"	"		"	-4.0mA							"	Y6	"		"		"		"
		32	"	"	"	"	"	"	-4.0mA	"								"	Y7	"		"		"		"
V_{OL6}	3007	33	"	"	"	0.8V	"	2.0V		"							20µA	4.5V	Y0		0.05		0.05		0.05	"
		34	2.0V	"	"	"	"	"		"						20µA		"	Y1		"		"		"	"
		35	0.8V	2.0V	"	"	"	"		"					20µA			"	Y2		"		"		"	"
		36	2.0V	2.0V	"	"	"	"		"				20µA				"	Y3		"		"		"	"
		37	0.8V	0.8V	2.0V	"	"	"		"			20µA					"	Y4		"		"		"	"
		38	2.0V	0.8V	"	"	"	"		"		20µA						"	Y5		"		"		"	"
		39	0.8V	2.0V	"	"	"	"		"	20µA							"	Y6		"		"		"	"
		40	2.0V	2.0V	"	"	"	"	20µA	"								"	Y7		"		"		"	"
V_{OL7}	3007	41	0.8V	0.8V	0.8V	0.8V	0.8V	2.0V		"						4.0mA	4.0mA	5.5V	Y0		0.26		0.4		0.26	"
		42	2.0V	0.8V	"	"	"	"		"					4.0mA			"	Y1		"		"		"	"
		43	0.8V	2.0V	"	"	"	"		"				4.0mA				"	Y2		"		"		"	"
		44	2.0V	2.0V	"	"	"	"		"			4.0mA					"	Y3		"		"		"	"
		45	0.8V	0.8V	2.0V	"	"	"		"		4.0mA						"	Y4		"		"		"	"
		46	2.0V	0.8V	"	"	"	"		"								"	Y5		"		"		"	"
		47	0.8V	2.0V	"	"	"	"		"	4.0mA							"	Y6		"		"		"	"
		48	2.0V	2.0V	"	"	"	"	4.0mA	"								"	Y7		"		"		"	"

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Case 2 / E and F / Test no.	Terminal conditions 1/ 2 / 1 / A	3 / 2 / B	4 / 3 / C	5 / 4 / $\overline{G2A}$	7 / 5 / $\overline{G2B}$	8 / 6 / G1	9 / 7 / Y7	10 / 8 / GND	12 / 9 / Y6	13 / 10 / Y5	14 / 11 / Y4	15 / 12 / Y3	17 / 13 / Y2	18 / 14 / Y1	19 / 15 / Y0	20 / 16 / V_{CC}	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
I_{OS4}	3011	49	GND	GND	GND	4.0V	4.0V	GND		GND								4.0V	Y0	-10	-120	-10	-120	-10	-120	mA
		50	"	"	"	"	"	"		"						GND		"	Y1	"	"	"	"	"	"	"
		51	"	"	"	"	"	"		"					GND		GND	"	Y2	"	"	"	"	"	"	"
		52	"	"	"	"	"	"		"				GND				"	Y3	"	"	"	"	"	"	"
		53	"	"	"	"	"	"		"			GND					"	Y4	"	"	"	"	"	"	"
		54	"	"	"	"	"	"		"		GND						"	Y5	"	"	"	"	"	"	"
		55	"	"	"	"	"	"		"	GND							"	Y6	"	"	"	"	"	"	"
		56	"	"	"	"	"	"	GND	"								"	Y7	"	"	"	"	"	"	"
I_{IH}	3010	57	5.5V	"	"	GND	GND	"		"								5.5V	A		0.05		0.1			µA
		58	GND	5.5V	"	"	"	"		"								"	B		"		"			"
		59	"	GND	5.5V	"	"	"		"								"	C		"		"			"
		60	"	"	GND	5.5V	"	"		"								"	$\overline{G2A}$		"		"			"
		61	"	"	"	GND	5.5V	"		"								"	G2B		"		"			"
		62	"	"	"	GND	GND	5.5V		"								"	G1		"		"			"
I_{IL}	3009	63	"	5.5V	5.5V	5.5V	5.5V	"		"								"	A		-0.05		-0.1			"
		64	5.5V	GND	5.5V	"	"	"		"								"	B		"		"			"
		65	"	5.5V	GND	"	"	"		"								"	C		"		"			"
		66	"	"	5.5V	GND	"	"		"								"	$\overline{G2A}$		"		"			"
		67	"	"	"	5.5V	GND	"		"								"	G2B		"		"			"
		68	"	"	"	5.5V	5.5V	GND		"								"	G1		"		"			"
																				Subgroup 4 T_C = +25°C Min	Subgroup 4 T_C = +25°C Max					
C_i	3012	69	2/							GND								GND	A		10					pF
		70		2/						"								"	B		"					"
		71			2/					"								"	C		"					"
		72				2/				"								"	$\overline{G2A}$		"					"
		73					2/			"								"	G2B		"					"
		74						2/		"								"	G1		"					"
																				Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 8 T_C = -55°C Min	Subgroup 8 T_C = -55°C Max	
Truth table tests 3/	3014	75	A	A	A	A	A	A	H	GND	H	H	H	H	H	H	H	4.5V	4/							
		76	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"								
		77	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"								
		78	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"								
		79	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"								
		80	"	"	"	A	B	"	"	"	"	"	"	"	"	"	"	"	"								
		81	"	"	"	A	B	B	"	"	"	"	"	"	"	"	"	"	"								
		82	"	"	"	B	A	"	"	"	"	"	"	"	"	"	"	"	"								
		83	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"								
		84	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	L	"								
		85	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	H	"								
		86	"	"	"	"	"	"	"	"	"	"	"	"	"	L	H	"	"								
		87	"	"	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"								
		88	"	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"	"								
		89	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"	"	"								
		90	"	"	"	"	"	"	"	"	"	L	H	"	"	"	"	"	"								
		91	"	"	"	"	"	"	L	"	L	H	"	"	"	"	"	"	"								

See footnotes at end of table.

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL1}	3003 (Fig. 4)	92	IN	GND	GND	GND	GND	4.5V		GND							OUT	4.5V	A to Y0	6	35	6	47	6	35	ns
t_{PHL1}		93	GND	IN	GND	"	"	"		"							"	"	B to Y0	"	"	"	"	"	"	"
t_{PHL1}		94	"	GND	IN	"	"	"		"							"	"	C to Y0	"	"	"	"	"	"	"
t_{PHL2}		95	"	"	GND	IN	"	"		"							"	"	$\overline{G2A}$ to Y0	4	"	4	"	4	"	"
t_{PHL2}		96	"	"	"	GND	IN	"		"							"	"	G2B to Y0	"	"	"	"	"	"	"
t_{PHL2}		97	"	"	"	"	GND	IN		"							"	"	G1 to Y0	"	"	"	"	"	"	"
t_{PHL1}		98	IN	"	"	"	"	4.5V		"						OUT		"	A to Y1	6	"	6	"	6	"	"
t_{PHL1}		99	4.5V	IN	"	"	"	"		"						"		"	B to Y1	"	"	"	"	"	"	"
t_{PHL1}		100	"	GND	IN	"	"	"		"						"		"	C to Y1	"	"	"	"	"	"	"
t_{PHL2}		101	"	"	GND	IN	"	"		"						"		"	$\overline{G2A}$ to Y1	4	"	4	"	4	"	"
t_{PHL2}		102	"	"	"	GND	IN	"		"						"		"	G2B to Y1	"	"	"	"	"	"	"
t_{PHL2}		103	"	"	"	"	GND	IN		"						"		"	G1 to Y1	"	"	"	"	"	"	"
t_{PHL1}		104	IN	4.5V	"	"	"	4.5V		"					OUT			"	A to Y2	6	"	6	"	6	"	"
t_{PHL1}		105	GND	IN	"	"	"	"		"					"			"	B to Y2	"	"	"	"	"	"	"
t_{PHL1}		106	"	4.5V	IN	"	"	"		"					"			"	C to Y2	"	"	"	"	"	"	"
t_{PHL2}		107	"	"	GND	IN	"	"		"					"			"	$\overline{G2A}$ to Y2	4	"	4	"	4	"	"
t_{PHL2}		108	"	"	"	GND	IN	"		"					"			"	G2B to Y2	"	"	"	"	"	"	"
t_{PHL2}		109	"	"	"	"	GND	IN		"					"			"	G1 to Y2	"	"	"	"	"	"	"
t_{PHL1}		110	IN	"	"	"	"	4.5V		"				OUT				"	A to Y3	6	"	6	"	6	"	"
t_{PHL1}		111	4.5V	IN	"	"	"	"		"				"				"	B to Y3	"	"	"	"	"	"	"
t_{PHL1}		112	"	4.5V	IN	"	"	"		"				"				"	C to Y3	"	"	"	"	"	"	"
t_{PHL2}		113	"	"	"	IN	"	"		"				"				"	$\overline{G2A}$ to Y3	4	"	4	"	4	"	"
t_{PHL2}		114	"	"	"	GND	IN	"		"				"				"	G2B to Y3	"	"	"	"	"	"	"
t_{PHL2}		115	"	"	"	"	GND	IN		"				"				"	G1 to Y3	"	"	"	"	"	"	"
t_{PHL1}		116	IN	GND	4.5V	"	"	4.5V		"			OUT					"	A to Y4	6	"	6	"	6	"	"
t_{PHL1}		117	GND	IN	4.5V	"	"	"		"			"					"	B to Y4	"	"	"	"	"	"	"
t_{PHL1}		118	"	GND	IN	"	"	"		"			"					"	C to Y4	"	"	"	"	"	"	"
t_{PHL2}		119	"	"	4.5V	IN	"	"		"			"					"	$\overline{G2A}$ to Y4	4	"	4	"	4	"	"
t_{PHL2}		120	"	"	"	GND	IN	"		"			"					"	G2B to Y4	"	"	"	"	"	"	"
t_{PHL2}		121	"	"	"	"	GND	IN		"			"					"	G1 to Y4	"	"	"	"	"	"	"
t_{PHL1}		122	IN	GND	"	"	"	4.5V		"		OUT						"	A to Y5	6	"	6	"	6	"	"
t_{PHL1}		123	4.5V	IN	"	"	"	"		"		"						"	B to Y5	"	"	"	"	"	"	"
t_{PHL1}		124	"	GND	IN	"	"	"		"		"						"	C to Y5	"	"	"	"	"	"	"
t_{PHL2}		125	"	"	4.5V	IN	"	"		"		"						"	$\overline{G2A}$ to Y5	4	"	4	"	4	"	"
t_{PHL2}		126	"	"	"	GND	IN	"		"		"						"	G2B to Y5	"	"	"	"	"	"	"
t_{PHL2}		127	"	"	"	"	GND	IN		"		"						"	G1 to Y5	"	"	"	"	"	"	"
t_{PHL1}		128	IN	4.5V	"	"	"	4.5V		"	OUT							"	A to Y6	6	"	6	"	6	"	"
t_{PHL1}		129	GND	IN	"	"	"	"		"	"							"	B to Y6	"	"	"	"	"	"	"
t_{PHL1}		130	"	4.5V	IN	"	"	"		"	"							"	C to Y6	"	"	"	"	"	"	"
t_{PHL2}		131	"	"	4.5V	IN	"	"		"	"							"	$\overline{G2A}$ to Y6	4	"	4	"	4	"	"
t_{PHL2}		132	"	"	"	GND	IN	"		"	"							"	G2B to Y6	"	"	"	"	"	"	"
t_{PHL2}		133	"	"	"	"	GND	IN		"	"							"	G1 to Y6	"	"	"	"	"	"	"
t_{PHL1}		134	IN	"	"	"	"	4.5V	OUT	"								"	A to Y7	6	"	6	"	6	"	"
t_{PHL1}		135	4.5V	IN	"	"	"	"	"	"								"	B to Y7	"	"	"	"	"	"	"
t_{PHL1}		136	"	4.5V	IN	"	"	"	"	"								"	C to Y7	"	"	"	"	"	"	"
t_{PHL2}		137	"	"	4.5V	IN	"	"	"	"								"	$\overline{G2A}$ to Y7	4	"	4	"	4	"	"
t_{PHL2}		138	"	"	"	GND	IN	"	"	"								"	G2B to Y7	"	"	"	"	"	"	"
t_{PHL2}		139	"	"	"	"	GND	IN	"	"								"	G1 to Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PLH1}	3003 (Fig. 4)	140	IN	GND	GND	GND	GND	4.5V		GND							OUT	4.5V	A to Y0	5	35	5	47	5	35	ns
t_{PLH1}		141	GND	IN	GND	"	"	"		"							"	"	B to Y0	"	"	"	"	"	"	"
t_{PLH1}		142	"	GND	IN	"	"	"		"							"	"	C to Y0	"	"	"	"	"	"	"
t_{PLH2}		143	"	"	GND	IN	"	"		"							"	"	$\overline{G2A}$ to Y0	4	"	4	"	4	"	"
t_{PLH2}		144	"	"	"	GND	IN	"		"							"	"	G2B to Y0	"	"	"	"	"	"	"
t_{PLH2}		145	"	"	"	"	GND	IN		"							"	"	G1 to Y0	"	"	"	"	"	"	"
t_{PLH1}		146	IN	"	"	"	"	4.5V		"						OUT		"	A to Y1	5	"	5	"	5	"	"
t_{PLH1}		147	4.5V	IN	"	"	"	"		"						"		"	B to Y1	"	"	"	"	"	"	"
t_{PLH1}		148	"	GND	IN	"	"	"		"						"		"	C to Y1	"	"	"	"	"	"	"
t_{PLH2}		149	"	"	GND	IN	"	"		"						"		"	$\overline{G2A}$ to Y1	4	"	4	"	4	"	"
t_{PLH2}		150	"	"	"	GND	IN	"		"						"		"	G2B to Y1	"	"	"	"	"	"	"
t_{PLH2}		151	"	"	"	"	GND	IN		"						"		"	G1 to Y1	"	"	"	"	"	"	"
t_{PLH1}		152	IN	4.5V	"	"	"	4.5V		"					OUT			"	A to Y2	5	"	5	"	5	"	"
t_{PLH1}		153	GND	IN	"	"	"	"		"					"			"	B to Y2	"	"	"	"	"	"	"
t_{PLH1}		154	"	4.5V	IN	"	"	"		"					"			"	C to Y2	"	"	"	"	"	"	"
t_{PLH2}		155	"	"	GND	IN	"	"		"					"			"	$\overline{G2A}$ to Y2	4	"	4	"	4	"	"
t_{PLH2}		156	"	"	"	GND	IN	"		"					"			"	G2B to Y2	"	"	"	"	"	"	"
t_{PLH2}		157	"	"	"	"	GND	IN		"					"			"	G1 to Y2	"	"	"	"	"	"	"
t_{PLH1}		158	IN	"	"	"	"	4.5V		"				OUT				"	A to Y3	5	"	5	"	5	"	"
t_{PLH1}		159	4.5V	IN	"	"	"	"		"				"				"	B to Y3	"	"	"	"	"	"	"
t_{PLH1}		160	"	4.5V	IN	"	"	"		"				"				"	C to Y3	"	"	"	"	"	"	"
t_{PLH2}		161	"	"	GND	IN	"	"		"				"				"	$\overline{G2A}$ to Y3	4	"	4	"	4	"	"
t_{PLH2}		162	"	"	"	GND	IN	"		"				"				"	G2B to Y3	"	"	"	"	"	"	"
t_{PLH2}		163	"	"	"	"	GND	IN		"				"				"	G1 to Y3	"	"	"	"	"	"	"
t_{PLH1}		164	IN	GND	4.5V	"	"	4.5V		"			OUT					"	A to Y4	5	"	5	"	5	"	"
t_{PLH1}		165	GND	IN	4.5V	"	"	"		"			"					"	B to Y4	"	"	"	"	"	"	"
t_{PLH1}		166	"	GND	IN	"	"	"		"			"					"	C to Y4	"	"	"	"	"	"	"
t_{PLH2}		167	"	"	4.5V	IN	"	"		"			"					"	$\overline{G2A}$ to Y4	4	"	4	"	4	"	"
t_{PLH2}		168	"	"	"	GND	IN	"		"			"					"	G2B to Y4	"	"	"	"	"	"	"
t_{PLH2}		169	"	"	"	"	GND	IN		"			"					"	G1 to Y4	"	"	"	"	"	"	"
t_{PLH1}		170	IN	GND	"	"	"	4.5V		"		OUT						"	A to Y5	5	"	5	"	5	"	"
t_{PLH1}		171	4.5V	IN	"	"	"	"		"		"						"	B to Y5	"	"	"	"	"	"	"
t_{PLH1}		172	"	GND	IN	"	"	"		"		"						"	C to Y5	"	"	"	"	"	"	"
t_{PLH2}		173	"	"	4.5V	IN	"	"		"		"						"	$\overline{G2A}$ to Y5	4	"	4	"	4	"	"
t_{PLH2}		174	"	"	"	GND	IN	"		"		"						"	G2B to Y5	"	"	"	"	"	"	"
t_{PLH2}		175	"	"	"	"	GND	IN		"		"						"	G1 to Y5	"	"	"	"	"	"	"
t_{PLH1}		176	IN	4.5V	"	"	"	4.5V		"	OUT							"	A to Y6	5	"	5	"	5	"	"
t_{PLH1}		177	GND	IN	"	"	"	"		"	"							"	B to Y6	"	"	"	"	"	"	"
t_{PLH1}		178	"	4.5V	IN	"	"	"		"	"							"	C to Y6	"	"	"	"	"	"	"
t_{PLH2}		179	"	"	4.5V	IN	"	"		"	"							"	$\overline{G2A}$ to Y6	4	"	4	"	4	"	"
t_{PLH2}		180	"	"	"	GND	IN	"		"	"							"	G2B to Y6	"	"	"	"	"	"	"
t_{PLH2}		181	"	"	"	"	GND	IN		"	"							"	G1 to Y6	"	"	"	"	"	"	"
t_{PLH1}		182	IN	"	"	"	"	4.5V	OUT	"								"	A to Y7	5	"	5	"	5	"	"
t_{PLH1}		183	4.5V	IN	"	"	"	"	"	"								"	B to Y7	"	"	"	"	"	"	"
t_{PLH1}		184	"	4.5V	IN	"	"	"	"	"								"	C to Y7	"	"	"	"	"	"	"
t_{PLH2}		185	"	"	4.5V	IN	"	"	"	"								"	$\overline{G2A}$ to Y7	4	"	4	"	4	"	"
t_{PLH2}		186	"	"	"	GND	IN	"	"	"								"	$\overline{G2B}$ to Y7	"	"	"	"	"	"	"
t_{PLH2}		187	"	"	"	"	GND	IN	"	"								"	G1 to Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	A	B	C	$\overline{G2A}$	$\overline{G2B}$	G1	Y7	GND	Y6	Y5	Y4	Y3	Y2	Y1	Y0	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{THL}	3004 (Fig. 4)	188	GND	GND	GND	GND	GND	IN		GND							OUT	4.5V	Y0	3	15	3	20	3	15	ns
		189	4.5V	GND	"	"	"	"		"						OUT		"	Y1	"	"	"	"	"	"	"
		190	GND	4.5V	"	"	"	"		"					OUT			"	Y2	"	"	"	"	"	"	"
		191	4.5V	4.5V	"	"	"	"		"				OUT				"	Y3	"	"	"	"	"	"	"
		192	GND	GND	4.5V	"	"	"		"			OUT					"	Y4	"	"	"	"	"	"	"
		193	4.5V	GND	"	"	"	"		"		OUT						"	Y5	"	"	"	"	"	"	"
		194	GND	4.5V	"	"	"	"		"	OUT							"	Y6	"	"	"	"	"	"	"
		195	4.5V	4.5V	"	"	"	"	OUT	"								"	Y7	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 4)	196	GND	GND	GND	"	"	"		"							OUT	"	Y0	"	"	"	"	"	"	"
		197	4.5V	GND	"	"	"	"		"						OUT		"	Y1	"	"	"	"	"	"	"
		198	GND	4.5V	"	"	"	"		"					OUT			"	Y2	"	"	"	"	"	"	"
		199	4.5V	4.5V	"	"	"	"		"				OUT				"	Y3	"	"	"	"	"	"	"
		200	GND	GND	4.5V	"	"	"		"			OUT					"	Y4	"	"	"	"	"	"	"
		201	4.5V	GND	"	"	"	"		"		OUT						"	Y5	"	"	"	"	"	"	"
		202	GND	4.5V	"	"	"	"		"	OUT							"	Y6	"	"	"	"	"	"	"
		203	4.5V	4.5V	"	"	"	"	OUT	"								"	Y7	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 53.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA							1/								GND	G1		1.5					V
		2		1mA						"								"	A1		"					"
		3			1mA					"								"	B1		"					"
		4								"							1mA	"	G2		"					"
		5								"						1mA		"	A2		"					"
		6								"					1mA			"	B2		"					"
V_{IC}(neg)		7	-1mA							GND								1/	G1		-1.5					"
		8		-1mA						"								"	A1		"					"
		9			-1mA					"								"	B1		"					"
		10								"							-1mA	"	G2		"					"
		11								"						-1mA		"	A2		"					"
		12								"					-1mA			"	B2		"					"
I_{CC}	3005	13	5.5V	5.5V	5.5V					"					5.5V	5.5V	5.5V	5.5V	V_{CC}		0.2		20			µA
$I_{CC}\Delta$		14	2.4V	0.8V						"					0.8V	0.8V	5.5V	"	V_{CC}		.345		1.5			mA
		15	0.8V	0.8V						"					0.8V	2.4V	GND	"	V_{CC}		.345		1.5			mA
V_{OH6}	3006	16	2.0V	0.8V	0.8V	-20µA				"					GND	GND		4.5V	1Y0	4.40		4.40		4.40		V
		17	"	"	"		-20µA			"					"	"		"	1Y1	"		"		"		"
		18	"	"	"			-20µA		"					"	"		"	1Y2	"		"		"		"
		19	"	"	"				-20µA	"					"	"		"	1Y3	"		"		"		"
		20		"	"					"				-20µA	"	"	2.0V	"	2Y0	"		"		"		"
		21		"	"					"			-20µA		"	"	"	"	2Y1	"		"		"		"
		22		"	"					"		-20µA			"	"	"	"	2Y2	"		"		"		"
		23		"	"					"	-20µA				"	"	"	"	2Y3	"		"		"		"
V_{OH7}	3006	24	2.0V	"	"	-4.0mA				"					"	"		5.5V	1Y0	3.98		3.70		3.98		"
		25	"	"	"		-4.0mA			"					"	"		"	1Y1	"		"		"		"
		26	"	"	"			-4.0mA		"					"	"		"	1Y2	"		"		"		"
		27	"	"	"				-4.0mA	"					"	"		"	1Y3	"		"		"		"
		28		"	"					"				-4.0mA	"	"	2.0V	"	2Y0	"		"		"		"
		29		"	"					"			-4.0mA		"	"	"	"	2Y1	"		"		"		"
		30		"	"					"		-4.0mA			"	"	"	"	2Y2	"		"		"		"
		31		"	"					"	-4.0mA				"	"	"	"	2Y3	"		"		"		"
V_{OL6}	3007	32	0.8V	"	"	20µA				"					0.8V	0.8V	0.8V	4.5V	1Y0		0.05		0.05		0.05	"
		33	"	2.0V	"		20µA			"					"	"	"	"	1Y1		"		"		"	"
		34	"	0.8V	2.0V			20µA		"					"	"	"	"	1Y2		"		"		"	"
		35	"	2.0V	2.0V				20µA	"					"	"	"	"	1Y3		"		"		"	"
		36	"	0.8V	0.8V					"				20µA	"	"	"	"	2Y0		"		"		"	"
		37	"	"	"					"			20µA		"	2.0V	"	"	2Y1		"		"		"	"
		38	"	"	"					"		20µA			2.0V	0.8V	"	"	2Y2		"		"		"	"
		39	"	"	"					"	20µA				2.0V	2.0V	"	"	2Y3		"		"		"	"
V_{OL7}	3007	40	"	"	"	4.0mA				"					0.8V	0.8V	"	4.5V	1Y0		0.26		0.4		0.26	"
		41	"	2.0V	"		4.0mA			"					"	"	"	"	1Y1		"		"		"	"
		42	"	0.8V	2.0V			4.0mA		"					"	"	"	"	1Y2		"		"		"	"
		43	"	2.0V	2.0V				4.0mA	"					"	"	"	"	1Y3		"		"		"	"
		44	"	0.8V	0.8V					"				4.0mA	"	"	"	"	2Y0		"		"		"	"
		45	"	"	"					"			4.0mA		"	2.0V	"	"	2Y1		"		"		"	"
		46	"	"	"					"		4.0mA			2.0V	0.8V	"	"	2Y2		"		"		"	"
		47	"	"	"					"	4.0mA				2.0V	2.0V	"	"	2Y3		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Case 2 / E and F / Test no.	Terminal conditions 1/ 2 / 1 / G1	3 / 2 / A1	4 / 3 / B1	5 / 4 / 1Y0	7 / 5 / 1Y1	8 / 6 / 1Y2	9 / 7 / 1Y3	10 / 8 / GND	12 / 9 / 2Y3	13 / 10 / 2Y2	14 / 11 / 2Y1	15 / 12 / 2Y0	17 / 13 / B2	18 / 14 / A2	19 / 15 / G2	20 / 16 / V_{CC}	Measured terminal	Test limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
I_{OS4}	3011	48	4.0V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND		4.0V	1Y0	-10	-120	-10	-120	-10	-120	mA
		49	"	"	"					"					"	"		"	1Y1	"	"	"	"	"	"	"
		50	"	"	"					"					"	"		"	1Y2	"	"	"	"	"	"	"
		51	"	"	"					"					"	"		"	1Y3	"	"	"	"	"	"	"
		52		"	"					"					"	"	4.0V	"	2Y0	"	"	"	"	"	"	"
		53		"	"					"			GND		"	"	"	"	2Y1	"	"	"	"	"	"	"
		54		"	"					"		GND			"	"	"	"	2Y2	"	"	"	"	"	"	"
		55		"	"					"	GND				"	"	"	"	2Y3	"	"	"	"	"	"	"
I_{IH}	3010	56	5.5V	"	"					"					"	"	GND	5.5V	G1		0.05		0.1			µA
		57	GND	5.5V	"					"					"	"	"	"	A1		"		"			"
		58	"	GND	5.5V					"					"	"		"	B1		"		"			"
		59	"	"	GND					"					"	"	5.5V	"	G2		"		"			"
		60	"	"	"					"					"	5.5V	GND	"	A2		"		"			"
		61	"	"	"					"					5.5V	GND	GND	"	B2		"		"			"
I_{IL}	3009	62	GND	5.5V	5.5V					"					"	5.5V	5.5V	"	G1		-0.05		-0.1			µA
		63	5.5V	GND	5.5V					"					"	"	"	"	A1		"		"			"
		64	"	5.5V	GND					"					"	"		"	B1		"		"			"
		65	"	"	5.5V					"					"	"	GND	"	G2		"		"			"
		66	"	"	"					"					"	GND	5.5V	"	A2		"		"			"
		67	"	"	"					"					GND	5.5V	5.5V	"	B2		"		"			"
																				Subgroup 4 T_C = +25°C Min	Subgroup 4 T_C = +25°C Max					
C_i	3012	68	2/	2/	2/					GND					2/	2/	2/	GND	G1	10						pF
		69								"								"	A1	"						"
		70								"								"	B1	"						"
		71								"								"	G2	"						"
		72								"								"	A2	"						"
		73								"								"	B2	"						"
																				Subgroup 7 T_C = +25°C Min	Subgroup 7 T_C = +25°C Max	Subgroup 8 T_C = +125°C Min	Subgroup 8 T_C = +125°C Max	Subgroup 8 T_C = -55°C Min	Subgroup 8 T_C = -55°C Max	
Truth table tests 3/	3014	74	A	A	A	H	H	H	H	GND	H	H	H	H	A	A	A	4.5V	4/							
		75	"	B	A	"	"	"	"	"	"	"	"	"	A	B	"	"									
		76	"	A	B	"	"	"	"	"	"	"	"	"	B	A	"	"									
		77	"	B	"	"	"	"	"	"	"	"	"	"	"	B	"	"									
		78	B	B	"	L	"	"	"	"	"	"	"	L	"	B	B	"									
		79	"	A	"	H	L	"	"	"	"	"	L	H	"	A	"	"									
		80	"	B	A	"	H	L	"	"	"	L	H	"	A	B	"	"									
		81	"	A	A	"	H	H	L	"	L	H	H	"	A	A	"	"									

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Unit
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Subgroup 9 T_C = +25°C 5/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 5/		
		E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
		Test no.	G1	A1	B1	1Y0	1Y1	1Y2	1Y3	GND	2Y3	2Y2	2Y1	2Y0	B2	A2	G2	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHL2}	3003 (Fig. 4)	82	IN	GND	GND	OUT				GND					GND	GND	GND	4.5V	G1 to 1Y0	5	35	5	47	5	35	ns
t_{PHL1}		83	GND	IN	GND	"				"					"	"	"	"	A1 to 1Y0	"	31	"	41	"	31	"
t_{PHL1}		84	GND	GND	IN	"				"					"	"	"	"	B1 to 1Y0	"	31	"	41	"	31	"
t_{PHL2}		85	IN	4.5V	GND		OUT			"					"	"	"	"	G1 to 1Y1	"	35	"	47	"	35	"
t_{PHL1}		86	GND	IN	GND		"			"					"	"	"	"	A1 to 1Y1	"	31	"	41	"	31	"
t_{PHL1}		87	GND	4.5V	IN		"			"					"	"	"	"	B1 to 1Y1	"	31	"	41	"	31	"
t_{PHL2}		88	IN	GND	4.5V			OUT		"					"	"	"	"	G1 to 1Y2	"	35	"	47	"	35	"
t_{PHL1}		89	GND	IN	4.5V			"		"					"	"	"	"	A1 to 1Y2	"	31	"	41	"	31	"
t_{PHL1}		90	GND	GND	IN			"		"					"	"	"	"	B1 to 1Y2	"	31	"	41	"	31	"
t_{PHL2}		91	IN	4.5V	4.5V				OUT	"					"	"	"	"	G1 to 1Y3	"	35	"	47	"	35	"
t_{PHL1}		92	GND	IN	4.5V				"	"					"	"	"	"	A1 to 1Y3	7	31	7	41	7	31	"
t_{PHL1}		93	"	4.5V	IN				"	"					"	"	"	"	B1 to 1Y3	7	31	7	41	7	31	"
t_{PHL2}		94	"	GND	GND					"				OUT	"	"	IN	"	G2 to 2Y0	5	35	5	47	5	35	"
t_{PHL1}		95	"	"	"					"				"	"	IN	GND	"	A2 to 2Y0	"	31	"	41	"	31	"
t_{PHL1}		96	"	"	"					"				"	IN	GND	GND	"	B2 to 2Y0	"	31	"	41	"	31	"
t_{PHL2}		97	"	"	"					"			OUT		GND	4.5V	IN	"	G2 to 2Y1	"	35	"	47	"	35	"
t_{PHL1}		98	"	"	"					"			"		GND	IN	GND	"	A2 to 2Y1	"	31	"	41	"	31	"
t_{PHL1}		99	"	"	"					"			"		IN	4.5V	GND	"	B2 to 2Y1	"	31	"	41	"	31	"
t_{PHL2}		100	"	"	"					"		OUT			4.5V	GND	IN	"	G2 to 2Y2	"	35	"	47	"	35	"
t_{PHL1}		101	"	"	"					"		"			4.5V	IN	GND	"	A2 to 2Y2	"	31	"	41	"	31	"
t_{PHL1}		102	"	"	"					"		"			IN	GND	GND	"	B2 to 2Y2	"	31	"	41	"	31	"
t_{PHL2}		103	"	"	"					"	OUT				4.5V	4.5V	IN	"	G2 to 2Y3	"	35	"	47	"	35	"
t_{PHL1}		104	"	"	"					"	"				4.5V	IN	GND	"	A2 to 2Y3	7	31	7	41	7	31	"
t_{PHL1}		105	"	"	"					"	"				IN	4.5V	"	"	B2 to 2Y3	7	31	7	41	7	31	"
t_{PHL2}		106	IN	"	"	OUT				"					GND	GND	"	"	G1 to 1Y0	5	35	5	47	5	35	"
t_{PHL1}		107	GND	IN	"	"				"					"	"	"	"	A1 to 1Y0	"	31	"	41	"	31	"
t_{PHL1}		108	GND	GND	IN	"				"					"	"	"	"	B1 to 1Y0	"	31	"	41	"	31	"
t_{PHL2}		109	IN	4.5V	GND		OUT			"					"	"	"	"	G1 to 1Y1	"	35	"	47	"	35	"
t_{PHL1}		110	GND	IN	GND		"			"					"	"	"	"	A1 to 1Y1	"	31	"	41	"	31	"
t_{PHL1}		111	GND	4.5V	IN		"			"					"	"	"	"	B1 to 1Y1	"	31	"	41	"	31	"
t_{PHL2}		112	IN	GND	4.5V			OUT		"					"	"	"	"	G1 to 1Y2	"	35	"	47	"	35	"
t_{PHL1}		113	GND	IN	4.5V			"		"					"	"	"	"	A1 to 1Y2	"	31	"	41	"	31	"
t_{PHL1}		114	GND	GND	IN			"		"					"	"	"	"	B1 to 1Y2	"	31	"	41	"	31	"
t_{PHL2}		115	IN	4.5V	4.5V				OUT	"					"	"	"	"	G1 to 1Y3	"	35	"	47	"	35	"
t_{PHL1}		116	GND	IN	4.5V				"	"					"	"	"	"	A1 to 1Y3	7	31	7	41	7	31	"
t_{PHL1}		117	"	GND	IN				"	"					"	"	"	"	B1 to 1Y3	7	31	7	41	7	31	"
t_{PHL2}		118	"	"	GND					"				OUT	"	"	IN	"	G2 to 2Y0	5	35	5	47	5	35	"
t_{PHL1}		119	"	"	"					"				"	"	IN	GND	"	A2 to 2Y0	"	31	"	41	"	31	"
t_{PHL1}		120	"	"	"					"				"	IN	GND	GND	"	B2 to 2Y0	"	31	"	41	"	31	"
t_{PHL2}		121	"	"	"					"			OUT		GND	4.5V	IN	"	G2 to 2Y1	"	35	"	47	"	35	"
t_{PHL1}		122	"	"	"					"			"		GND	IN	GND	"	A2 to 2Y1	"	31	"	41	"	31	"
t_{PHL1}		123	"	"	"					"			"		IN	4.5V	GND	"	B2 to 2Y1	"	31	"	41	"	31	"
t_{PHL2}		124	"	"	"					"		OUT			4.5V	GND	IN	"	G2 to 2Y2	"	35	"	47	"	35	"
t_{PHL1}		125	"	"	"					"		"			4.5V	IN	GND	"	A2 to 2Y2	"	31	"	41	"	31	"
t_{PHL1}		126	"	"	"					"		"			IN	GND	GND	"	B2 to 2Y2	"	31	"	41	"	31	"
t_{PHL2}		127	"	"	"					"	OUT				4.5V	4.5V	IN	"	G2 to 2Y3	"	35	"	47	"	35	"
t_{PHL1}		128	"	"	"					"	"				4.5V	IN	GND	"	A2 to 2Y3	7	31	7	41	7	31	"
t_{PHL1}		129	"	"	"					"	"				IN	4.5V	GND	"	B2 to 2Y3	7	31	7	41	7	31	"

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TABLE III. Group A inspection for device type 53 – Continued.

Symbol	MIL-STD-883 method	Case 2 / E and F / Test no.	Terminal conditions 1/ 2 / 1 / G1	3 / 2 / A1	4 / 3 / B1	5 / 4 / 1Y0	7 / 5 / 1Y1	8 / 6 / 1Y2	9 / 7 / 1Y3	10 / 8 / GND	12 / 9 / 2Y3	13 / 10 / 2Y2	14 / 11 / 2Y1	15 / 12 / 2Y0	17 / 13 / B2	18 / 14 / A2	19 / 15 / G2	20 / 16 / V_{CC}	Measured terminal	Test limits: Subgroup 9 T_C = +25°C 5/ Min	Subgroup 9 Max	Subgroup 10 T_C = +125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C 5/ Min	Subgroup 11 Max	Unit
t_{THL}	3003 (Fig. 4)	130	IN	GND	GND					GND					GND	GND	GND	4.5V	1Y0	3	15	3	20	3	15	ns
		131	"	4.5V	GND					"					"	"	"	"	1Y1	"	"	"	"	"	"	"
		132	"	GND	4.5V					"					"	"	"	"	1Y2	"	"	"	"	"	"	"
		133	"	4.5V	4.5V					"					"	"	"	"	1Y3	"	"	"	"	"	"	"
		134	GND	GND	GND					"					"	"	IN	"	2Y0	"	"	"	"	"	"	"
		135	"	"	"					"					"	4.5V	"	"	2Y1	"	"	"	"	"	"	"
		136	"	"	"					"					4.5V	GND	"	"	2Y2	"	"	"	"	"	"	"
		137	"	"	"					"					4.5V	4.5V	"	"	2Y3	"	"	"	"	"	"	"
t_{TLH}	3003 (Fig. 4)	138	IN	"	"					"					GND	GND	GND	"	1Y0	"	"	"	"	"	"	"
		139	"	4.5V	"					"					"	"	"	"	1Y1	"	"	"	"	"	"	"
		140	"	GND	4.5V					"					"	"	"	"	1Y2	"	"	"	"	"	"	"
		141	"	4.5V	4.5V					"					"	"	"	"	1Y3	"	"	"	"	"	"	"
		142	GND	GND	GND					"					"	"	IN	"	2Y0	"	"	"	"	"	"	"
		143	"	"	"					"					"	4.5V	"	"	2Y1	"	"	"	"	"	"	"
		144	"	"	"					"					4.5V	GND	"	"	2Y2	"	"	"	"	"	"	"
		145	"	"	"					"					4.5V	4.5V	"	"	2Y3	"	"	"	"	"	"	"

1/ Output pins not designated may be high level logic, low level logic, open. Input pins not designated may be high level logic or low level logic. Exceptions are as follows:
   a. V_{IC}(pos) tests, the GND terminal shall be open. A minimum limit of 0.4 V applies to tests being performed on equipment not capable of opening the GND pin during test.
   b. V_{IC}(neg) tests, the V_{CC} terminal shall be open.
   c. I_{CC} tests, the output terminal shall be open.

2/ See 4.4.1c.

3/ Tests shall be performed in sequence. A = 2.4 V; B = 0.4 V for device types 52 and 53; A = 3.7 V; B = 0.4 V for device types 01 through 04.

4/ X = indeterminate output voltage. Output shall be H > 2.5 V; L < 2.5 V.

5/ See 4.4.1d.

6/ The input signal for HCT device types shall be 0 to 3 volts; however, the 50 percent V_{CC} measuring point is 1.3 V for input and output signals. The 10 percent and 90 percent V_{CC} points are 0.3 V and 2.7 V, respectively for HCT devices.

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4.5 Methods of inspection. Methods of inspection shall be specified and as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit GND terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

Parameter 1/	All device types
I_{CC}	±30 nA

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5.3 Quiescent supply current (I_{CC} test). When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow through the meter.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements are as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance and radiation hardness assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

j. Packaging requirements. (see 5.1)

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6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractors parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, P.O. Box 3990, Columbus, Ohio 43218-3990.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

C_i ..... Input terminal-to-GND capacitance.
I_{CC} ..... Quiescent supply current.
$I_{CC}\Delta$ ..... Additional supply current when using worst case TTL input levels.
T_C ..... Case temperature.
GND ..... Ground zero voltage potential.
V_{CC} ..... Positive supply voltage.

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, will be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests (see 3.6).

b. A copy of each radiograph.

c. The technology conformance inspection (TCI) data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.6).

e. Final electrical parameters data (see 4.2d).

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	54HC42
02	54HC138
03	54HC139
04	54HC154
52	54HCT138
53	54HCT139

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6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-2005-024)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force – 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at http://assist.daps.dla.mil/.

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